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In Partial Fulfillment of the Requirements in

IT 302 - System Integration and Architecture

MIDTERM PROJECT

MASTER TEST PLAN

of

Clinic Teleconsultation Booking and Reminder System

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I. Test Plan Identifier

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Version	Date	Description of Changes	Prepared By
0.1	September 16, 2026	Initial draft for static review; review issues identified	Maranan, Aryana Rizanne N.
1.0	September 29, 2026	Static-review findings corrected; exit criteria, severity scale, and monitoring added; RTM and Appendices C, D, and E rebuilt to the Guide templates; PR-09 and ER-06 to ER-08 added; peer review comments from Tamayo, Jonie R. (QA Engineer) incorporated.	Maranan, Aryana Rizanne N.
1.0 (restructured)	October 1, 2026	Reorganized into the Master Test Plan format with Component, Integration, System, and Acceptance Test Plans.	Maranan, Aryana Rizanne N.

II. References

Ref. No.	Document Title	Document Link
REF-01	Assigned Scenario D - Clinic Teleconsultation Booking and Reminder	IT302_Midterm_Project_Master_Test_Plan_Studen.pdf
REF-02	IT302 Midterm Project Student Guide (MTP Format and Rubric; 19-component Master Test Plan format)	IT302_Midterm_Project_Master_Test_Plan_Studen.pdf
REF-03	IEEE Std 829-2008 - Standard for Software and System Test Documentation	IEEE SA - IEEE 829-2008
REF-04	ISTQB CTFL Syllabus v4.0.1 (Sec. 1.3 testing principles, Ch. 5 test management, Sec. 5.1.7 test quadrants)	https://www.istqb.org/wp-content/uploads/2024/11/ISTQB_CTFL_Syllabus_v4.0.1.pdf
REF-05	ISTQB CTAL-AT Syllabus v2.0 (2026)	https://istqb.org/certifications/certified-tester-advanced-level-agile-tester-ctal-at/
REF-06	MediLynk Activity 1 System Blueprint (architecture, technology stack, feature list)	MARANAN, Aryana Rizanne N._SIA_ACT 1 (1).pdf
REF-07	IT302 System Integration and Architecture Course Syllabus, First Semester, SY 2026-2027	IT302-System Integration and Architecture with Automation_CourseSyllabus.pdf

Ref. No.	Document Title	Document Link
REF-08	Possible Testing per Scenario and Recommended Database (Scenarios A-F: scope, testing ideas, data flow, main tables)	Possible Testing per Scenario and Recommended Database.pdf
REF-09	ISO/IEC/IEEE 29119-3 - Software testing, Part 3: Test documentation	https://standards.ieee.org/ieee/29119-3/7499
REF-10	Appendix C - Early Testing / Static Review Log (this document)	Part of the test basis (clarified requirements)

III. Introduction

Purpose

MediLynk Solutions is a web and mobile-based Clinic Teleconsultation Booking and Reminder System with Automated Patient Notification Management. Following the three-tier architecture in the Activity 1 System Blueprint (REF-06), the presentation tier is a React web application and a Flutter mobile application; the logic tier is a Node.js/Express server using the MVC pattern, with the Observer pattern driving reminder notifications and the Singleton pattern managing shared resources such as the database connection; and the data tier is a MySQL database (Users, Doctors & Schedules, Appointments, Consultations, Medical Records, Notification Logs).

The system has four roles: Patient, Doctor/Healthcare Provider, Clinic Admin/Staff, and System Administrator (a platform-level role distinct from clinic-level Admin/Staff). It integrates with external Email, SMS, and Push notification providers through REST APIs.

The purpose of this Master Test Plan (MTP) is to ensure the reliability, functionality, and performance of MediLynk before it enters operational use, and to document how the group will plan, statically review, risk-assess, and design the test strategy. The plan is risk-based: the highest-scored product risks (double booking, unauthorized access, and calendar synchronization failure) receive the earliest and deepest testing, which directly supports risk mitigation.

Scope

This MTP covers the full set of testing levels - Component, Integration, System, and Acceptance - for ten requirements (REQ-01 to REQ-10) and sixteen test items (TI-01 to TI-16) of the MediLynk Teleconsultation System. The functional areas covered are appointment booking and slot-conflict checking, role-based access, calendar synchronization, automated reminders, data masking, reschedule/cancel transitions, service recovery, teleconsultation video sessions, patient registration validation, and expired-slot booking. Standards used as structural references are IEEE 829-2008, ISO/IEC/IEEE 29119-3, and the ISTQB syllabi (REF-03 to REF-05, REF-09).

This Midterm Project covers planning, static/early testing, risk analysis, and test-strategy design only. Full dynamic execution, production-scale load testing, and Selenium automation are out of scope (see Section VII).

Objective

The objective of this MTP is to outline the procedure the team will follow when testing the system, and to confirm that it meets the project/business requirements and quality standards. Specifically, the testing effort aims to:

1. Verify that booking, rescheduling, and slot-conflict logic rejects double booking, schedule overlap, and expired-slot bookings for every doctor, including the required buffer time of 10 minutes between consecutive appointments (ER-06). (REQ-01, REQ-10, PR-01, PR-07)
2. Verify that role-based access control and data masking restrict confidential patient health information according to the four roles, and that patient registration rejects empty, invalid, and boundary data. (REQ-02, REQ-05, REQ-09, PR-02, PR-04, PR-06)

3. Verify that the calendar updates correctly after booking, rescheduling, and cancelling, and that failed synchronization is retried and flagged. (REQ-03, REQ-06, PR-05, PR-08)
4. Verify that automated reminders are triggered at the required lead times and delivered through the confirmed channel(s) via the Notification Manager. (REQ-04, PR-03)
5. Verify that a teleconsultation video session starts within 10 seconds under normal conditions, and that a documented recovery option (Rejoin option, then a reschedule prompt) is shown when the video platform is unavailable or a call drops. (REQ-07, REQ-08, PR-09)

Assumptions

- The course reference sheet (REF-08) recommends PostgreSQL for Scenario D. The Activity 1 blueprint (REF-06) selected MySQL, which is retained as an equivalent technically sound choice: MySQL with the InnoDB engine supports transactions, foreign keys, and composite unique constraints (doctor + time slot) for PR-01. MySQL has no native row-level security, so REQ-02 and REQ-05 are enforced at the service layer (TI-01, TI-11) and by role-restricted database accounts or views, and are verified by the role x data decision-table and OWASP ZAP tests rather than by database policies.
- Sandbox or trial credentials for the Email API, SMS Gateway, and Push Notification Service, plus a mock video-conferencing simulator, will be available for planning-phase verification.
- Test accounts for all four roles will be provisioned before dynamic execution begins.
- The clarifications in Appendix C are the agreed test basis for this Midterm scope. The thresholds in ER-06 to ER-08 are team-defined values pending instructor confirmation, and the role x field matrix (ER-08) remains an open residual risk.

- The architecture, technology stack, and feature list in the Activity 1 System Blueprint remain the agreed system design for this Midterm scope.

Constraints

- The group has a two-week planning window and no access to real (non-synthetic) patient data because of data-privacy constraints.
- Free-tier/sandbox quotas of the Email, SMS, and Push providers may limit the number of test notifications (see TR-02).

IV. Test Items

This test plan covers a set of essential elements for the MediLynk Teleconsultation System, including software products, source code, and third-party apps. The team carefully identified these requirements from the master test plan, detailed documentation, and other relevant resources. The purpose of testing these components is to ensure that what is delivered meets specified requirements and maintains high-quality standards. Requirement, risk, priority, quadrant, and test-condition traceability is in Sections V and VI and Appendices A and B.

TI-01 – Authentication Module

Handles user login and issues JWT-based sessions to keep users securely authenticated. It also enforces role-based access control (RBAC) so each user only reaches what their role permits.

TI-02 – Appointment / Schedule Management Module

Manages appointment booking and the calendar logic that organizes doctor

availability. Its conflict-check engine prevents double-bookings and overlapping schedules.

TI-03 – Symptom & File Management Module

Provides the intake form where patients describe their symptoms before a consultation. It also supports uploading and attaching supporting files such as photos or lab results.

TI-04 – Teleconsultation Management Module

Manages the video and call sessions between patients and doctors. It handles session setup, connection, and completion of each remote consultation.

TI-05 – User Management Module

Manages the creation, updating, and maintenance of Patient, Doctor, and Staff accounts. It keeps user profiles and account statuses organized and consistent across the system.

TI-06 – Notification Manager

Uses the Observer pattern to trigger notifications automatically when system events occur. It dispatches messages through Email, SMS, and Push channels.

TI-07 – Medical Records Management Module

Stores and organizes clinical notes, patient history, and prescriptions. It gives authorized users secure access to a complete record of each patient's care.

TI-08 – Reporting Module

Provides dashboards that summarize system activity and key metrics for Admin and Staff. These reports support monitoring, decision-making, and operational oversight.

TI-09 – MySQL Database

Serves as the central data store for Users, Doctors & Schedules, Appointments, Consultations, Medical Records, and Notification Logs. It ensures data is stored reliably and remains consistent across all modules.

TI-10 – Third-Party Service Integrations

Connects the system to external services, including an Email API, SMS Gateway, and Push Notification Service. These integrations allow the platform to deliver messages without building its own delivery infrastructure.

TI-11 – RBAC Enforcement

Applies access control rules consistently across all modules for the four defined user roles. This ensures users can only view and perform actions that match their assigned permissions.

TI-12 – Service Recovery / Fallback Handler

Provides recovery logic for teleconsultation sessions that are interrupted by connection loss or technical failure. It helps restore the session or switch to a fallback option so care is not disrupted.

TI-13 – User Interface / Screens

Delivers role-based dashboards and views for both web and mobile platforms. Each role sees a tailored interface that supports its specific tasks and workflows.

TI-14 – Internal API Layer

Exposes the REST endpoints that connect the client applications to the backend services. It covers internal communication only, since third-party APIs are handled under TI-10.

TI-15 – Inter-Module Integration / Data Flow

Coordinates how data moves between modules, such as Appointment to Notification, Appointment/Schedule to the calendar service, and Teleconsultation to Medical Records. This keeps related modules synchronized so that one action correctly triggers the next.

TI-16 – Data Input/Output & Export Functions

Handles form submissions, file uploads and downloads, and data exports. It ensures data entering and leaving the system is processed correctly and in usable formats.

Test Cases

Test cases are designed from the test conditions (TCND-01 to TCND-14) in Appendix B, using the techniques listed in Section VIII (BVA, EP, decision table, state transition, error guessing, exploratory, and checklist-based). Each test level is owned by one lead member:

Component - Aduna and Estolloso; Integration - Estolloso; System - Marasigan (technical/security) with Aduna (functional); Acceptance - Tamayo and Estolloso. Every test case carries a Test Condition ID that maps to a requirement and product risk in the Requirements Traceability Matrix (RTM, Appendix B), ensuring bidirectional traceability.

Test Condition	**Requirement**	**Risk**	**Level**
TCND-01, 02	REQ-01	PR-01	Component / Integration / System
TCND-03	REQ-02	PR-02	Component / Integration / System
TCND-04, 05	REQ-03	PR-05	Integration
TCND-06, 14	REQ-04	PR-03	Integration / System
TCND-07	REQ-05	PR-06	System
TCND-08	REQ-06	PR-08	Component / Integration
TCND-09	REQ-07	PR-09	System
TCND-10	REQ-08	PR-09	System
TCND-11	REQ-09	PR-04	Component / System
TCND-12	REQ-10	PR-07	Component / System
TCND-13	REQ-01, 03, 04, 09	PR-01, 03, 05	System / Acceptance

V. Software Risk Issues

In the dynamic landscape of software development, the team emphasizes the critical importance of identifying and addressing potential risks to ensure the success and dependability of the system. Product risks are scored as Risk Score = Likelihood x Impact (1-5 each). Levels: High (16-25), Medium (8-15), Low (1-7). Full descriptions, mitigations, and test responses are in Appendix A. This section presents the three High-level risks in the required Description / Impact / Mitigation / Contingency structure, followed by the summary of all nine product risks.

RISK ISSUE # 1: Double Booking of Doctor and Slot (PR-01, Score 20 / High)

Description:

Two patients may book the same doctor and time slot at the same moment, or a booking may violate the 10-minute buffer between consecutive appointments. A clinic schedule with conflicting appointments cannot be honored, so this is a serious scheduling risk.

Impact:

This specific issue poses the following impacts on the system when the event occurs:

1. **Conflicting Appointments:** Two patients may be scheduled with the same doctor at the same time, so one consultation cannot proceed as planned.
2. **Patient and Doctor Inconvenience:** Affected patients and doctors face delays, cancellations, and rescheduling.
3. **Loss of Trust:** Repeated scheduling errors can reduce patient and doctor confidence in the system.
4. **Staff Workload:** Clinic staff must spend extra time identifying and resolving conflicts manually.

5. **Data Inconsistency:** Conflicting records may lead to inconsistent data in the Appointments table.

Contingency Plan:

In the event of a double booking or buffer violation, the project team should perform the following measures:

Mitigation Measures:

- **Database Constraint:** Add a unique constraint on doctor + time slot in the InnoDB database (TI-09) so the database rejects duplicate bookings.
- **Conflict-Check Engine:** Enforce the slot-conflict and 10-minute buffer rules in the conflict-check engine (TI-02) before any booking is confirmed.
- **Concurrency Testing:** Run simultaneous booking requests using JMeter and Postman to verify that only one booking succeeds.
- **Boundary Testing:** Apply Boundary Value Analysis (BVA) at the buffer edge, at the component, integration, and system levels.

Contingency Measures:

- **Manual Rescheduling:** If a double booking is found after release, Clinic Admin/Staff manually reschedule one appointment and notify the affected patient.
- **Defect Handling:** Log the defect as Critical and fix it with first retest priority.

Incident Response Team:

Assign a dedicated team to respond to scheduling conflicts, ensuring the team is well-trained and prepared to act promptly when a double booking or buffer violation is found.

The Clinic Admin/Staff handle rescheduling and patient notification, while the backend developers fix the conflict-check engine (TI-02) and database constraint (TI-09). The QA team logs the defect as Critical and gives it first retest priority.

RISK ISSUE # 2: Unauthorized Access to Consultation History (PR-02, Score 20 / High)

Description:

A patient could view another patient's consultation history, or a doctor could view consultations not assigned to them, because of weak access rules. Consultation history is confidential health information, so any exposure is a serious concern.

Impact:

This specific issue poses the following impacts on the system when the event occurs:

1. **Data Breach:** Unauthorized users may gain access to sensitive patient consultation records.
2. **Privacy-Law Violations:** Exposure of health data may violate privacy-law and data-privacy requirements.
3. **Reputational Damage:** A privacy incident can harm the clinic's standing with patients and the public.
4. **Loss of Patient Trust:** Patients may stop using the system if they believe their health information is not protected.

Contingency Plan:

In the event of unauthorized access to consultation history, the project team should perform the following measures:

Mitigation Measures:

- **Role-Based Access Checks:** Enforce access checks at the service layer (TI-01, TI-11) so each role only reaches data it is permitted to see.
- **Decision Table:** Build a role x data decision table to define exactly which roles can access which data.
- **Security Testing:** Run OWASP ZAP scans and manual authorization tests for each role.
- **Cross-Patient Testing:** Use at least two patient accounts to confirm that one patient cannot access another's records.

Contingency Measures:

- **Immediate Containment:** Disable the affected endpoint or role permission right away.
- **Log Review:** Review access logs in Notification Logs and server logs to determine what was accessed.
- **Rule Correction:** Correct the faulty access rule and re-verify it.
- **Release Control:** Treat the defect as Critical; it blocks release until fixed or formally risk-accepted by the Test Manager.

Incident Response Team:

Assign a dedicated team to respond to unauthorized access incidents, ensuring the team is well-trained and prepared to contain and investigate a privacy breach promptly. The security lead and backend developers disable the affected endpoint or role permission, review the access logs, and correct the access rule (TI-01, TI-11). The QA team re-verifies the fix, and the Test Manager decides whether the release can proceed or the risk is formally accepted.

RISK ISSUE # 3: Calendar Synchronization Failure (PR-05, Score 16 / High)

Description:

Synchronization between the consultation/appointment service and the calendar service may fail, so doctor availability displays incorrectly. This can leave the calendar out of step with the actual appointment records.

Impact:

This specific issue poses the following impacts on the system when the event occurs:

1. **Stale Availability:** Patients and staff may see slots as open or taken when they are not.
2. **Unnoticed Double Booking:** Outdated availability can allow conflicting appointments to go undetected.
3. **Missed Consultations:** Doctors or patients may not be aware of changes, leading to missed sessions.
4. **Inconsistent Doctor Schedules:** The calendar and the Appointments data no longer match.

Contingency Plan:

In the event of a calendar synchronization failure, the project team should perform the following measures:

Mitigation Measures:

- **Retry Logic:** Add sync retry logic and flagging for manual review (ER-04) to recover from failed updates.

- **State-Transition Testing:** Test calendar updates after booking, reschedule, and cancel across the states pending, failed, retried, and flagged.
- **Exploratory Timing Sessions:** Run exploratory sessions to uncover timing-related sync issues.
- **Performance Check:** Verify that availability updates appear within 5 seconds.

Contingency Measures:

- **Queue and Retry:** Failed syncs are queued and retried automatically.
- **Manual Review:** Syncs that keep failing are flagged for Clinic Admin/Staff, who correct the calendar manually until the sync is repaired.

Incident Response Team:

Assign a dedicated team to respond to synchronization failures, ensuring the team is well-trained and prepared to restore calendar accuracy promptly. The backend and integration developers repair the sync service and review the retry and flagging logic (ER-04). The Clinic Admin/Staff correct the calendar manually for flagged syncs until the repair is complete, and the QA team confirms that availability updates within 5 seconds.

VI. Features To Be Tested

This section lists the features and functions of MediLynk that fall within the scope of this Midterm test plan:

1. **Appointment Booking and Slot-Conflict Check:** Verify that a second booking for the same doctor and time slot is rejected, including simultaneous requests. Ensure that consecutive appointments respect the 10-minute buffer time.
2. **Role-Based Access to Consultation History:** Ensure that a Doctor can view only the consultations assigned to them. Confirm that Clinic Admin/Staff and System Administrator access only what their role allows.
3. **Calendar Synchronization Between Services:** Validate that doctor availability updates within 5 seconds after a booking is made. Verify that a failed sync is retried and flagged for manual review.
4. **Automated Reminder Notifications:** Verify that reminders are sent 24 hours and 1 hour before the appointment, with a tolerance of 2 minutes and the correct date and time (time zones per ER-08). Ensure delivery works through the confirmed channels: Email, SMS, and Push.
5. **Data Masking of Confidential Health Information:** Ensure that sensitive fields are hidden by default from roles without clearance. Validate this behavior against the role x field matrix (ER-08).
6. **Reschedule and Cancel Appointment Status Transitions:** Verify that an appointment status moves only through valid states. Confirm that each transition frees or updates the doctor slot, leaving no conflicting or outdated slot.

7. **Service Recovery / Fallback Handler:** Test that the documented recovery option is shown to both Patient and Doctor when the video platform is unavailable or a call drops. Ensure that a Rejoin option appears first, followed by a reschedule prompt if rejoin fails within 2 minutes.
8. **Teleconsultation Video Session:** Validate that a session starts within 10 seconds of being initiated. Confirm that a 10-minute test session runs with no unplanned disconnection under normal conditions.
9. **Patient Registration Validation:** Verify that empty, invalid, and boundary values are rejected with clear error messages. Ensure that valid data is saved correctly.
10. **Expired Time-Slot Booking:** Ensure that booking a past or expired slot is rejected. Confirm that this is enforced through server-side date validation.

VII. Features Not To Be Tested

This section includes all features integrated in the system, but will not be included in the scope of testing:

1. **Full Dynamic Test Execution:** This Midterm covers planning, static review, risk analysis, and strategy design only, so full test execution is outside its scope.
2. **Production-Scale Load and Stress Testing:** No performance risk is above Medium, and the free-tier sandbox quotas (TR-02) prevent realistic volume. Small-scale concurrency (simultaneous booking requests, PR-01) and video-session performance under normal conditions remain in scope.
3. **Selenium/Playwright UI Automation:** Later automation is planned for the final project, so it is not required at Midterm.
4. **Internal Operation of Third-Party Email, SMS, and Push Services:** These services are the third party's responsibility. Only MediLynk's integration is tested (TI-10).
5. **Internal Operation of the Third-Party Video Platform:** The platform itself is the third party's responsibility. Only session handling and recovery are tested (TI-04, TI-12).
6. **Real (Non-Synthetic) Patient Data:** Data-privacy constraints rule out real patient data, so only synthetic data is used.
7. **Full UAT Execution:** Full UAT is not required at Midterm. It is planned in Appendix D (Q3) and in the Acceptance Test Plan.
8. **Reporting Module Dashboards (TI-08):** There is no identified risk or requirement (REQ-01 to REQ-10) for this module, so testing is deferred.

9. **Symptom & File Management (TI-03); Data Input/Output and Export (TI-16):** No requirement or risk falls within the Midterm scope for these modules, so testing is deferred.
10. **iOS Devices and Wider Mobile/Browser Matrix:** Only Chrome, Edge, and Android (device or emulator) are available, and iOS is not. Compatibility testing is limited to these.

VIII. Approach

In this master test plan, the project team decided to utilize the following approaches, which will cover the planning and testing phases of MediLynk. The overall strategy is risk-based: test effort, order, and technique follow the product-risk scores in Appendix A. Static and early testing (requirements review, testability review, risk analysis, and review of ambiguous or missing requirements and acceptance conditions, Appendix C) precedes dynamic testing. Static review sits outside the quadrant count.

These are the following Test Approaches:

1. **Risk-Based Testing**

   Risk-based testing prioritizes the testing effort according to the likelihood and impact of product failures. For MediLynk, the highest-scoring risks are concurrency, authorization, and synchronization (PR-01, PR-02, PR-05), so these areas receive the earliest and deepest coverage.

   The team tests the P1 risks (PR-01, PR-02, PR-05) first, then the P2 risks (starting with PR-06), and finally the P3 risks. This ordering keeps the effort aligned with business importance. It also ensures that the most damaging defects are found while there is still time to fix and retest them.

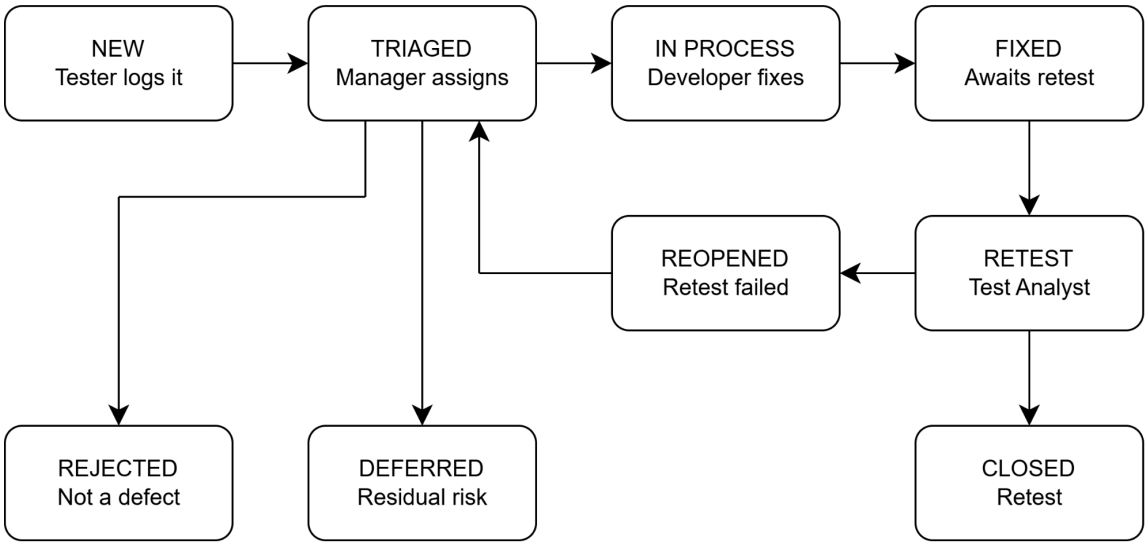


Figure 3. Defect Lifecycle Diagram

2. Component Testing

Component testing contributes to the overall reliability and stability of MediLynk by ensuring that each individual function works correctly within the larger system. It applies to the smallest testable parts, namely the TI-02 conflict-check engine, the TI-01/TI-11 RBAC checks, and the TI-05 validation logic.

These tests are fast and run early, and they are automated where feasible. Because they are performed first, defects in the core booking, access-control, and validation rules are found and corrected before the components are combined with others.

3. Integration and System Testing

Integration and system testing ensure that the components of MediLynk work together and that the complete system meets its requirements. Integration testing covers

the TI-15 inter-module flow, the TI-10 notification providers, and the TI-14 API layer, using Postman API checks and sandbox or mock services.

System testing then covers REQ-01 to REQ-10 end to end through a combination of manual testing and targeted automation. The test types applied are functional, security, usability, small-scale concurrency and performance, compatibility (Chrome, Edge, and Android), reliability, recovery, smoke, and exploratory testing.

4. **Acceptance Testing**

Acceptance testing is the final phase of testing. It validates whether the system meets the needs and expectations of its end users. At the Midterm stage, acceptance testing is for planning only (Appendix D, Q3), because full UAT execution is not required at this point.

Test Techniques

The team applies the following techniques to the areas where they are most effective.

- **Boundary Value Analysis (BVA):** Applied to PR-07, PR-04, the buffer-time edge of PR-01 (ER-06), and PR-03 (24 h / 1 h lead times, 2-minute tolerance), because date/time and input boundaries are exact failure points.
- **Equivalence Partitioning (EP):** Applied to PR-04 (registration fields) to group valid and invalid input classes for field validation.
- **Decision Table Testing:** Applied to PR-02 and PR-06, because outcomes depend on role x data-sensitivity combinations.

- **State Transition Testing:** Applied to PR-08 (REQ-06) and PR-05 (sync retry states), because appointment status and sync status (pending, failed, retried, flagged) move through defined states.
- **Error Guessing:** Applied to PR-01 (simultaneous booking) and PR-09 (simulated outage or dropped call), because concurrency and outage failures are surfaced by deliberately attempting conflicting or failing actions.
- **Exploratory Testing:** Applied to PR-03, PR-05 (timing sessions), and the video session (REQ-08), because failures depend on external gateway or service timing that is hard to script in advance.
- **Checklist-Based Testing:** Applied to REQ-08 session start and quality and to smoke checks, because a fixed list of verification points suits repeatable checks with no complex input combinations.

Testing Quadrants Summary (see Appendix D)

Q1 and Q4 carry the most weight because the highest risks are concurrency, authorization, and synchronization (PR-01, PR-02, PR-05). Q2 covers validation, reminders, and API behavior. Q3 is smaller at Midterm because execution is out of scope; it holds exploratory sessions, a usability walkthrough of booking and recovery screens, and UAT planning. No equal quota per quadrant is used.

Entry Criteria

Entry criteria apply to the dynamic execution phase that follows MTP approval. Before execution begins, the following conditions must be met:

- The test basis is reviewed, with high-risk ambiguities clarified or documented as residual risks.
- Accounts for all four roles are provisioned.
- Sandbox or mock services for Email/SMS/Push and the video simulator are reachable.
- Synthetic test data is loaded.
- Smoke checks pass on the build.

Monitoring and Control

The team tracks progress through the following planned metrics:

- Requirements coverage = REQs with at least one executed test condition / 10.
- Risk coverage = PRs with at least one executed test condition / 9.
- P1 test-condition status (passed / failed / blocked / not run).
- Progress against the schedule in Section XVI.
- Defects by severity and by test item (to detect clusters).
- Open blockers and defect age.

Progress is reported weekly. The following control actions apply when problems appear:

- A P1 failure gets first retest priority.
- A defect cluster in a test item triggers extra tests or an exploratory session in that area.

- A blocker triggers the suspension rules in Section X.
- A schedule slip is handled under TR-04 (reassign to a member with lighter workload).

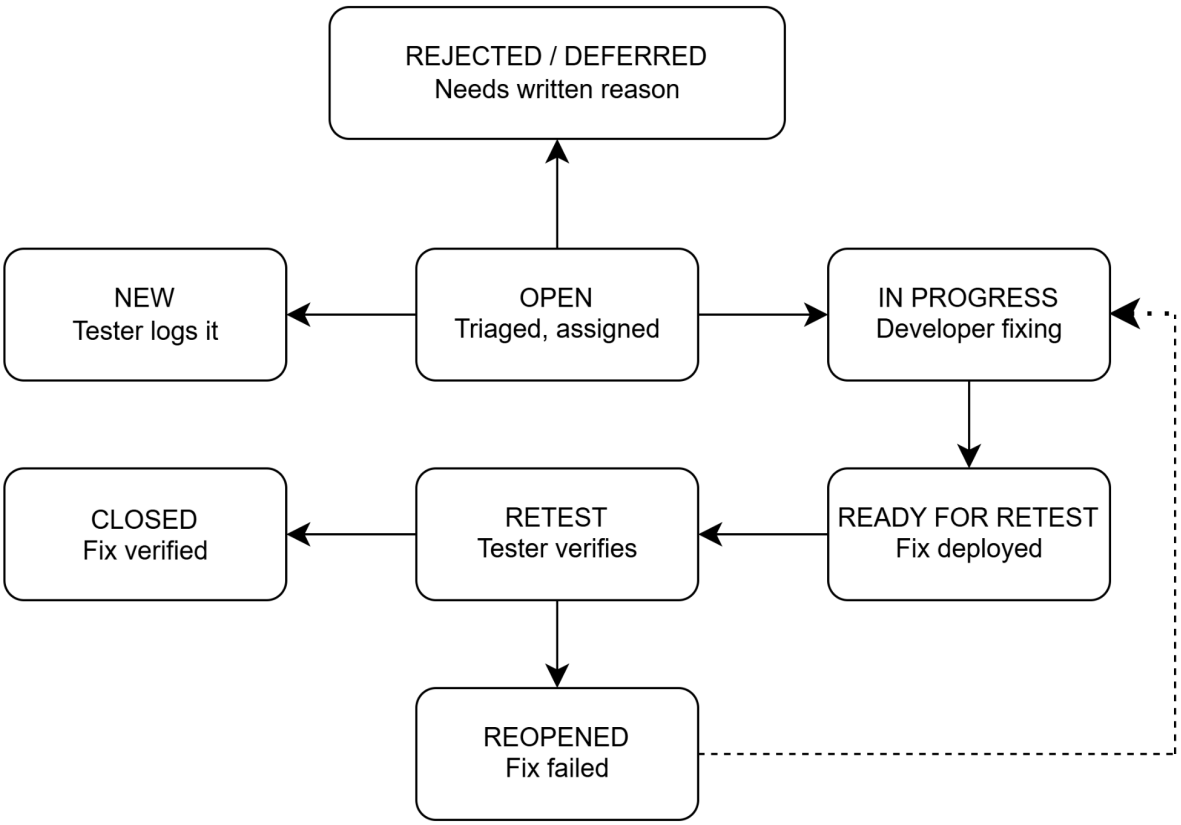


Figure 4. Defect Lifecycle in MediLynk Testing.

Test Tools

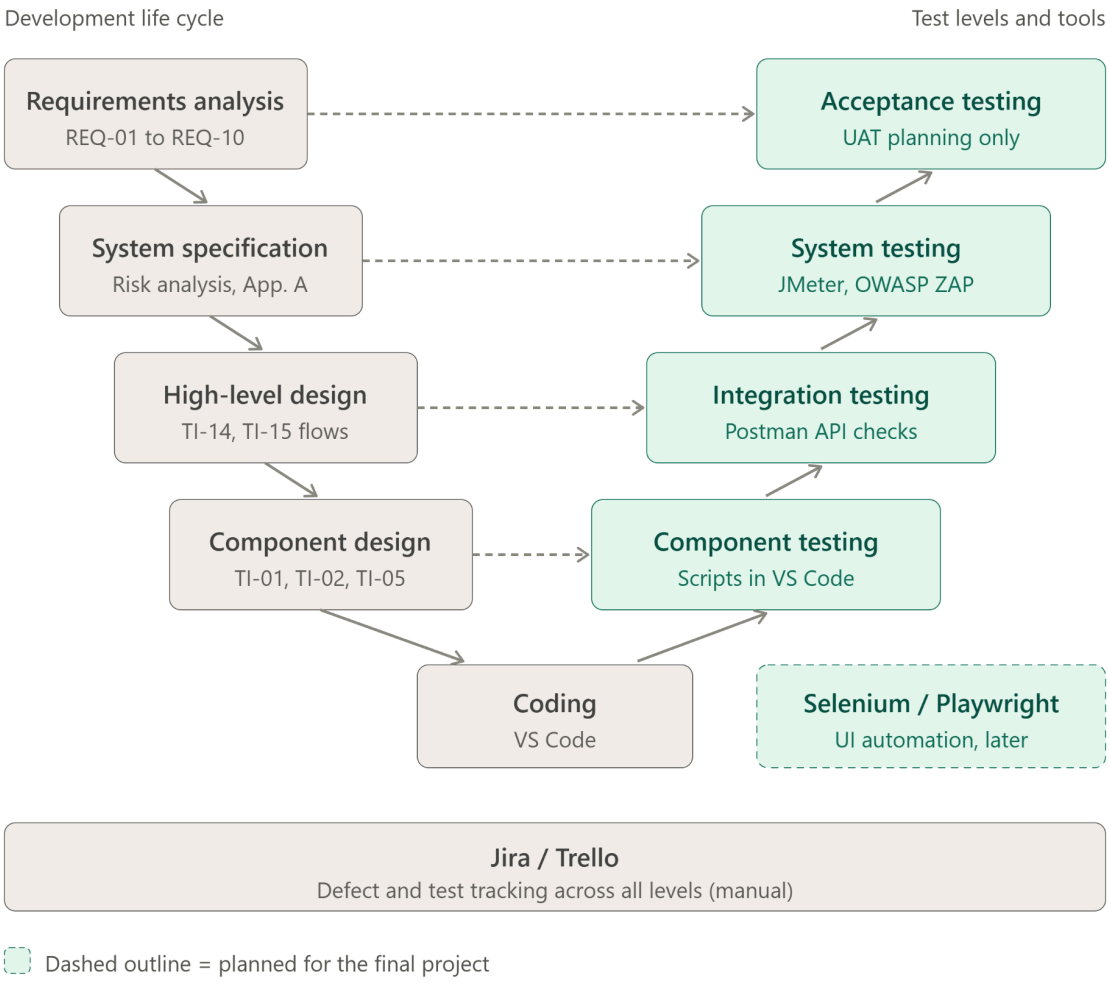


Figure 5. Systems Engineering Diagram of MediLynk Test Tools

The project team will use the V-model as the basis for assigning test tools across the MediLynk development process. The figure above shows each development stage on the left, matched to its test level on the right, with the tool used at that level. VS Code supports component testing, Postman supports integration testing, and JMeter and OWASP ZAP support system testing. Acceptance testing is for planning only at the Midterm, and Selenium /

Playwright is deferred to the final project. Jira / Trello tracks defects and tests progress across all levels.

Testing type	July				Aug				Sept				Oct	
Week	1	2	3	4	1	2	3	4	1	2	3	4	1	2
Component testing														
Integration testing														
System testing														
Acceptance testing														
Reviews														

Table 1. Gantt Chart

The table above shows the schedule for the MediLynk testing phases from July to October. Testing began in the third week of September, starting with component testing and moving through integration, system, and acceptance testing, with reviews closing out the schedule in October. The Gantt chart lets the team track the status of each phase against the planned weeks.

Testing type	Schedule
Component testing	Sept weeks 3 to 4
Integration testing	Sept week 4 to Oct week 1
System testing	Oct weeks 1 to 2
Acceptance testing	Oct week 2
Reviews	Oct week 2

IX. Item Pass/Fail Criteria

Functionality	Criteria	
	Pass	**Fail**
Booking and Slot Conflict (REQ-01, PR-01)	● For simultaneous requests on one doctor/slot, exactly one succeeds and the rest are rejected. ● A booking inside the 10-minute buffer is rejected.	● Two bookings are accepted for one slot. ● The buffer rule is violated.
Consultation History Access (REQ-02, PR-02)	● Each role can view only what its role allows. ● A Patient requesting another patient's history is denied. ● A Doctor requesting a consultation not assigned to them is denied.	● Any record is visible outside the role's permission. ● A Patient can view another patient's history. ● A Doctor can view a consultation not assigned to them.

Functionality	Criteria	
	Pass	**Fail**
Calendar Synchronization (REQ-03, PR-05)	● Availability is correct within 5 seconds of booking. ● A failed sync is retried and flagged.	● Availability is stale beyond 5 seconds. ● A failure is not retried or flagged.
Reminder Delivery (REQ-04, PR-03)	● Reminders are sent 24 h and 1 h before the appointment, within 2 minutes. ● Reminders show the correct date/time and channel.	● A reminder is late beyond the tolerance. ● A reminder is wrong or missing.
Data Masking (REQ-05, PR-06)	● Sensitive fields are hidden from every role without clearance, per the role x field matrix.	● Sensitive data is visible to an unauthorized role.
Reschedule/Cancel (REQ-06, PR-08)	● Only valid status transitions occur. ● The slot is updated within 5 seconds.	● An invalid transition is allowed. ● The calendar is outdated or conflicting.

Functionality	Criteria	
	Pass	**Fail**
Service Recovery (REQ-07, PR-09)	• On outage or dropped call, the documented recovery option (Rejoin option, then a reschedule prompt) is shown. • The option is shown to both Patient and Doctor.	• No recovery option is shown. • The option is shown to only one party.
Video Session (REQ-08, PR-09)	• The session starts within 10 seconds. • The 10-minute test session runs with no unplanned disconnection.	• The session fails to start. • The threshold is breached.
Patient Registration (REQ-09, PR-04)	• Invalid data is rejected with a clear error. • Valid data is saved correctly.	• Invalid data is accepted. • Valid data is not saved correctly.
Expired Slot Booking (REQ-10, PR-07)	• Booking of a past or expired slot is rejected by the server.	• Booking of an expired slot is accepted.

X. Suspension Criteria and Resumption Requirements

In case of any unwanted and critical scenarios that may affect the testing process of MediLynk, testing can be paused for the time needed for recovery. The following suspension criteria have been declared for this test plan:

1. **Blocking Defect in a High-Priority Risk:** If a High-priority risk (PR-01, PR-02, or PR-05) produces a blocking defect that prevents further execution of dependent test cases, testing will be suspended.
2. **Notification Gateway Unavailable:** If the SMS/email gateway (or its mock, per TR-02) is unavailable, reminder-related testing will be halted.
3. **Test Environment or Test Data Unavailable:** If the test environment or test data becomes unavailable, testing will be suspended.
4. **Unusable Build:** If the build is unusable because critical smoke checks fail, testing will be suspended.

Resumption Requirements:

Testing activities will only recommence once the following requirements are met:

1. **Blocking Defect Resolved:** The blocking defect is fixed and re-verified.

   Environment and Service Restored: The environment or service is restored and confirmed stable.
2. **Updated Build and Test Data:** The updated build and test data are delivered and smoke-tested.
3. **Test Manager Approval:** The Test Manager approves resumption.

4. **Mitigation Implemented:** The mitigation listed in Appendix A for the relevant risk has been implemented and confirmed.

XI. Test Deliverables

The test deliverables consist of documents and records that support the planning, execution, monitoring, and evaluation of testing activities for the Clinic Teleconsultation Booking and Reminder System. They provide evidence of test coverage, identified defects, and testing progress.

XI. Test Deliverables

The test deliverables consist of documents and records that support the planning, execution, monitoring, and evaluation of testing activities for the Clinic Teleconsultation Booking and Reminder System. They provide evidence of test coverage, identified defects, and testing progress. The owner of each deliverable is the role responsible for preparing it.

Test Plan

The master test plan (this document) and the four level-specific test plans that follow it are deliverables of this effort: the Component Test Plan, the Integration Test Plan, the System Test Plan, and the Acceptance Test Plan.

1. **Master Test Plan (MTP)** – Defines the overall testing objectives, scope, strategy, resources, schedule, risks, and responsibilities of the testing team. *Owner: Project Manager.*
    a. **Component, Integration, System, and Acceptance Test Plans** – Level-specific plans derived from the MTP. *Owner: Project Manager with level leads.*

2. **Product and Project/Test Risk Matrix** – Identifies product and project risks, their likelihood and impact, priority levels, and mitigation strategies. *Owner: Test Manager.*
3. **Preliminary Requirements Traceability Matrix (RTM)** – Connects requirements to risks, test conditions, testing levels, and test techniques. *Owner: Test Analyst.*
4. **Early Testing and Static Review Log** – Records issues found in reviewing requirements and workflows before execution. *Owner: QA Engineer.*
5. **Testing Quadrants Coverage Map** – Documents how testing activities cover the testing quadrants. *Owner: Technical Test Analyst.*
6. **Seven Testing Principles Evidence Matrix** – Describes how the seven testing principles are applied. *Owner: QA Engineer / Project Manager.*
7. **Test Conditions and Test Cases** – Conditions, procedures, expected results, and requirements used to verify functionality and integration. *Owner: Test Analyst.*
8. **Test Checklists** – Structured lists of verification points for booking, authentication, reminders, data validation, and other features. *Owner: Test Analyst.*
9. **Defect Reports** – Identified defects with description, severity, priority, reproduction steps, expected and actual results. *Owner: All testers.*

   Test Logs and Screenshots – Execution results with screenshots or other evidence. *Owner: Integration Test Engineer.*
10. **Progress and Status Reports** – Completed activities, current progress, issues, and remaining tasks. *Owner: Test Manager.*
11. **Test Completion Report** – Overall results, unresolved defects, achieved coverage, limitations, and recommendations. *Owner: Test Manager.*

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A.Y. 2026-2027

In Partial Fulfillment of the Requirements in
IT 302 - System Integration and Architecture

APPENDIX A - PRODUCT AND PROJECT/TEST RISK MATRIX

of

Clinic Teleconsultation Booking and Reminder System

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October 2026

APPENDIX A - PRODUCT AND PROJECT/TEST RISK MATRIX

I. Test Plan Identifier

CTP-IT302-MEDILYNK-1

II. References

Ref. No.	Document Title	Document Title
REF-MTP	MediLynk Master Test Plan v1.0	MediLynk Master Test Plan v1.0 (Sections VI, VIII, IX; Appendices A, B)
REF-06	MediLynk Activity 1 System Blueprint	MARANAN, Aryana Rizanne N._SIA_ACT 1 (1).pdf
REF-04	International Software Testing Qualifications Board (ISTQB). (2024). *Certified Tester Foundation Level Syllabus v4.0.1*.	ISTQB Certified Tester - Foundation Level Syllabus v4.0
REF-03	ISTQB CTFL Syllabus v4.0.1 (system testing, test quadrants)	IEEE SA - IEEE 829-2008

III. Introduction

Purpose

To verify that each individually testable unit of MediLynk - the conflict-check engine, authentication and RBAC checks, registration validation, status-transition logic, and the unique slot constraint - works correctly in isolation before modules are combined. Early component testing gives fast feedback on the highest-risk logic (PR-01, PR-02).

Scope

This component test plan covers the logic of the individual units of MediLynk, tested separately from the rest of the system. It covers the following:

- TI-02: the conflict engine, date validation, and status transitions.
- TI-01 / TI-11: authentication and RBAC checks.
- TI-05: registration validation.
- TI-09: the composite unique constraint.

Module-to-module data flow, third-party providers, and end-to-end scenarios are covered by the Integration and System Test Plans.

Objective

This component test plan specifically focuses on the following objectives:

- Requirements Coverage: To confirm that component-level behavior meets REQ-01, REQ-02, REQ-06, REQ-09, and REQ-10.
- Buffer Rule Enforcement: To confirm that the 10-minute buffer rule (ER-06) is enforced.

- Early Defect Removal: To ensure that defects in the P1 components are found and fixed before integration.

IV. Test Items

This test plan covers the following items of the MediLynk Teleconsultation System, identified from the Master Test Plan, the Activity 1 System Blueprint, and related documentation. Testing them ensures that what is delivered meets the specified requirements and maintains high-quality standards at the component level.

1. **Appointment / Schedule Management Module (TI-02)**

   This module contains the conflict engine, date validation, and status transitions, which form the core booking logic of the system. It is linked to REQ-01, REQ-06, and REQ-10.

2. **Authentication Module (TI-01)**

   This module handles login, JWT sessions, and RBAC checks, which control who can enter the system. It is linked to REQ-02.

3. **RBAC Enforcement (TI-11)**

   This item covers the service-layer permission checks that limit each role to the data and actions it is allowed. It is linked to REQ-02 and REQ-05.

4. **User Management Module (TI-05)**

This module handles registration validation for user accounts, so only complete and valid data is saved. It is linked to REQ-09.

5. **MySQL Database (TI-09)**

At this level, testing covers the unique constraint on doctor + time slot, which blocks duplicate bookings at the database. It is linked to REQ-01.

V. Software Risk Issues

The product risks below apply to this test level. Scores follow the MTP (Likelihood x Impact), and full descriptions are in Appendix A.

- **Double booking of doctor and slot (PR-01, 20 / High):** A unique constraint on doctor + time slot, the conflict-check engine, and concurrency tests address this risk.
- **Unauthorized access to consultation history (PR-02, 20 / High):** Service-layer RBAC checks, a role x data decision table, and security scans address this risk.
- **Incomplete patient registration data (PR-04, 6 / Low):** Field-level validation, EP, and BVA address this risk.
- **Booking of an expired time slot (PR-07, 4 / Low):** Server-side date validation and BVA on past and boundary dates address this risk.
- **Calendar not updated after reschedule/cancel (PR-08, 9 / Medium):** A calendar update on every reschedule/cancel and state consistency checks address this risk.

VI. Features To Be Tested

Component testing will focus on the following key features.

1. **Slot-Conflict Logic (REQ-01; TCND-01, 02)**
    - A second request for the same doctor/slot is rejected.
    - A booking within the 10-minute buffer is rejected.
    - The database unique constraint blocks duplicates.
2. **RBAC Checks (REQ-02; TCND-03)**
    - The permission function returns deny for cross-patient and unassigned-consultation access for each role.
3. **Status Transitions (REQ-06; TCND-08)**
    - Only valid appointment states are reachable.
    - Invalid transitions are refused.
4. **Registration Validation (REQ-09; TCND-11)**
    - Empty, invalid, and boundary values are rejected with clear error messages.
    - Valid data is saved.
5. **Expired-Slot Validation (REQ-10; TCND-12)**
    - Past and boundary-date slots are rejected by server-side date validation.

VII. Features Not To Be Tested

The following features will not be tested in this component testing phase.

1. **Inter-Module Data Flow, Calendar Sync, and Reminders**
    - These features are covered at the Integration level (TI-15).
2. **Video Sessions and Recovery (TI-04, TI-12)**
    - These features are covered at the System level.
3. **Third-Party Providers (TI-10)**
    - Third-party providers are the third party's responsibility, so only the integration is tested.
4. **Reporting, Symptom & File, and Export Modules**
    - These modules are excluded per MTP Section VII.

VIII. Approach

In the context of the MediLynk Teleconsultation System, the component testing approach involves an evaluation of individual components to verify their functionality and adherence to specific requirements. Each component is tested in isolation before it is combined with others, using stubs or mocks for the database and external services. In this way, defects in the booking, access-control, and validation logic can be found and corrected at an early stage, before integration and system testing.

Test Levels

Component (unit) testing is applied to individual functions and classes in isolation, using stubs or mocks for the database and external services. Direct SQL checks are also performed on the unique constraint.

Test Types

The following test types are applied at this level:

1. **Functional Testing**

   Functional testing verifies the business rules of the components, such as the slot-conflict logic and the status transitions, so that each behaves as specified.

2. **Security Testing**

   Security testing verifies the permission checks, so that each role can reach only the data and actions it is allowed.

3. **Boundary and Validation Testing**

   Boundary and validation testing verifies that input at the edges of valid ranges, as well as empty and invalid values, is handled correctly.

Test Techniques

The following techniques are used to design the test cases:

- Boundary Value Analysis (BVA): Applied to the buffer edge, past and boundary dates, and field limits.
- Equivalence Partitioning (EP): Applied to the registration input classes.
- Decision Table Testing: Applied to role x data permissions.
- State Transition Testing: Applied to the appointment status.
- Error Guessing: Applied to simultaneous inserts against the unique constraint.

Test Tools

The following tools support the component testing process. Each tool is matched to a specific purpose.

Tool	Purpose
Node.js unit-test framework (e.g., Jest; proposed, to be confirmed with the development team)	Automated unit tests of Express service functions
Postman	Direct calls to individual endpoints
MySQL client / VS Code	Constraint and data checks

IX. Item Pass/Fail Criteria

The following pass and fail criteria are designed to evaluate the component-level behavior of the MediLynk Teleconsultation System under test, ensuring that it meets the specified requirements. The criteria below determine whether each test item is working as intended at the component level.

Test Item	Pass Criteria	Fail Criteria
Conflict engine (REQ-01)	Exactly one of simultaneous requests succeeds; buffer violation rejected.	Two bookings accepted or buffer violated.
RBAC checks (REQ-02)	All role x data denials return deny.	Any unauthorized permission granted.
Status transitions (REQ-06)	Only valid transitions occur.	Invalid transition allowed.
Registration validation (REQ-09)	Invalid data rejected; valid saved.	Invalid data accepted.
Expired slot (REQ-10)	Past slot rejected.	Past slot accepted.

X. Suspension Criteria and Resumption Requirements

This section declares the suspension criteria and resumption requirements for the component testing of MediLynk. Testing can be paused for the time needed for recovery when any of the following occurs:

1. **Blocking Defect in a P1 Component:** A P1 component produces a blocking defect that prevents dependent tests.
2. **Build Failure:** The build fails to compile or start.
3. **Test Database or Test Data Unavailable:** The test database or test data becomes unavailable.

Resumption Requirements:

Testing activities will only recommence once the following requirements are met:

1. **Blocking Defect Resolved:** The blocking defect is fixed and re-verified.
2. **Environment Restored:** The environment or service is restored and confirmed stable.
3. **Updated Build and Test Data:** The updated build and test data are delivered and smoke-tested.
4. **Test Manager Approval:** The Test Manager approves resumption.
5. **Mitigation Confirmed:** The relevant Appendix A mitigation is confirmed.

XI. Test Deliverables

The following deliverables are produced during the component testing of MediLynk:

1. **Component Test Cases and Unit-Test Scripts:** The test cases and automated unit-test scripts used to verify each component.
2. **Test Logs and Defect Reports:** The execution results and the defects found during component testing.
3. **Component Test Summary:** The summary of component test results, which is included in the weekly progress report.

XII. Testing Task

The following testing tasks must be completed during component testing:

1. **Finalize Component Test Cases:** Finalize the component test cases from TCND-01, 02, 03, 08, 11, and 12. *Owner: Test Analyst.*
2. **Prepare Test Support Items:** Prepare the stubs, mocks, and synthetic data. *Owner: Integration Test Engineer.*
3. **Configure Framework and Database:** Configure the unit-test framework and the database. *Owner: Technical Test Analyst.*
4. **Execute Tests:** Execute the tests, with the P1 risks (PR-01, PR-02) first. *Owner: All testers.*
5. **Log and Retest Defects:** Log the defects found and retest them after the fixes. *Owner: Test Analyst.*

6. **Report Component Results:** Report the results of the component tests. *Owner: Test Manager.*

XIII. Environmental Needs

1. **Workstations:** Laptop or desktop with an Intel Core i5 or equivalent, 8 GB RAM, and 256 GB of free storage, running Windows 10/11.
2. **Browsers / Mobile:** Google Chrome and Microsoft Edge, plus an Android device or emulator. iOS is not available.
3. **Test Data:** Synthetic or anonymized data only, with accounts for Patient, Doctor, Clinic Admin/Staff, and System Administrator.
4. **Application Server:** The Node.js/Express (MVC) build under test.
5. **Database:** MySQL (InnoDB) with the six data-tier tables.

XIV. Responsibilities

In order to guarantee that component testing is carried out successfully, each member of the team is assigned a role with specific duties at this level.

- **Test Analyst – Aduna, Franz Juris U.:** Designs and executes the component tests for the business rules.
- **Integration Test Engineer – Estolloso, Cielo Marie S.:** Prepares the stubs and test data and executes the component tests.
- **Technical Test Analyst – Marasigan, Kylie Denise D.G.:** Configures the unit-test framework and the database.
- **Test Manager – Salvador, Mary Pauleen C.:** Monitor progress and priorities.

XV. Staffing and Training Needs

Component testing needs team members who can read Node.js code and write simple test scripts, so that each unit of MediLynk can be tested in isolation. The members assigned to each role are listed in Section XIV.

1. **Training for Testing Team:** Training refers to the learning opportunities given to the testing team to build the skills needed for component testing. The following training will be provided:
    - **Unit-Testing Framework Orientation:** Members learn how to set up the framework and write and run unit tests for the Express service functions.
    - **SQL Constraint Checks:** Members learn how to check the composite unique constraint and other data rules directly in the database.
    - **Test Case Design Workshop (BVA/EP):** Members practice designing test cases using Boundary Value Analysis and Equivalence Partitioning.

The following images show the assigned component testers watching training videos and learning how to perform their respective tasks.

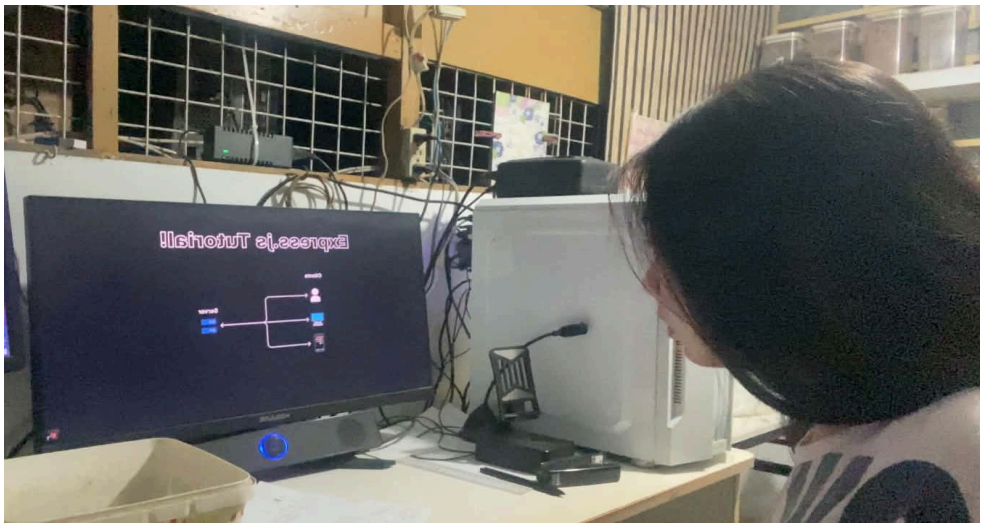


Figure 6. Unit-Testing Framework Orientation

The figure shows Kylie Denise D.G. Marasigan, the Technical Test Analyst, watching a Jest unit-testing tutorial.

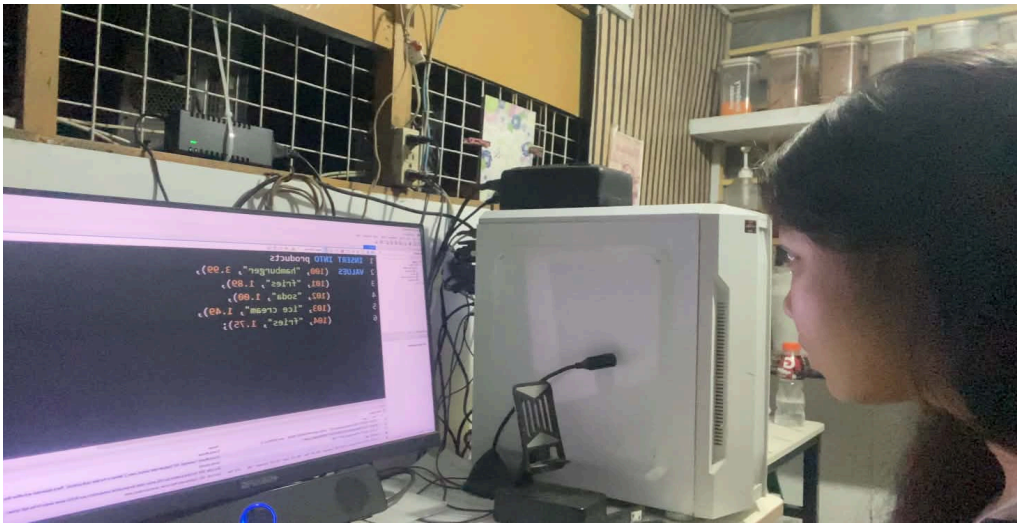


Figure 7. SQL Constraint Check

The figure shows Kylie Denise D.G. Marasigan, the Technical Test Analyst, watching a MySQL unique-constraint tutorial.

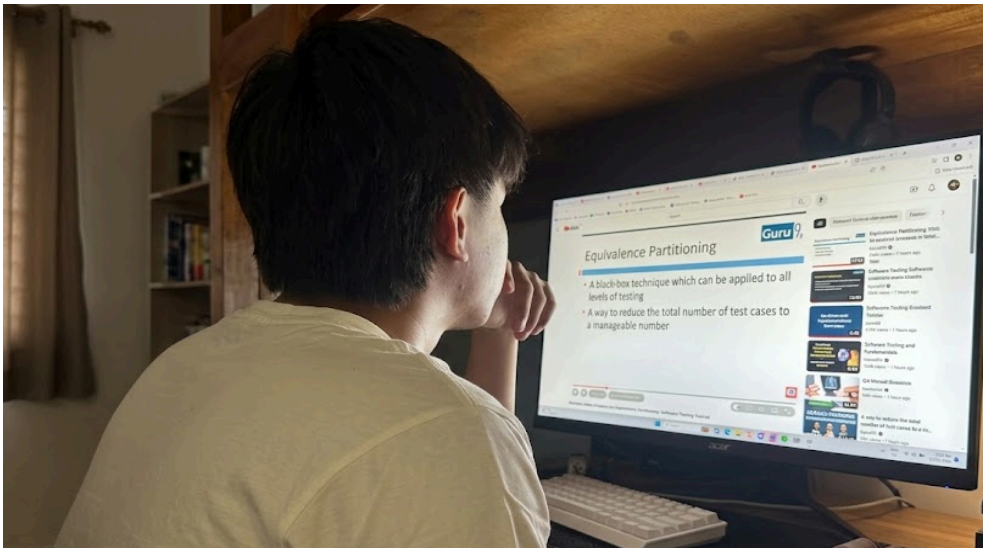


Figure 8. Test Case Design Workshop

The figure shows Franz Juris U. Aduna, the Test Analyst, and the testers watching a BVA and EP tutorial.

[screenshot]

Figure 9. Postman Endpoint Check

The figure shows Cielo Marie S. Estolloso, the Integration Test Engineer, watching a Postman API testing tutorial.

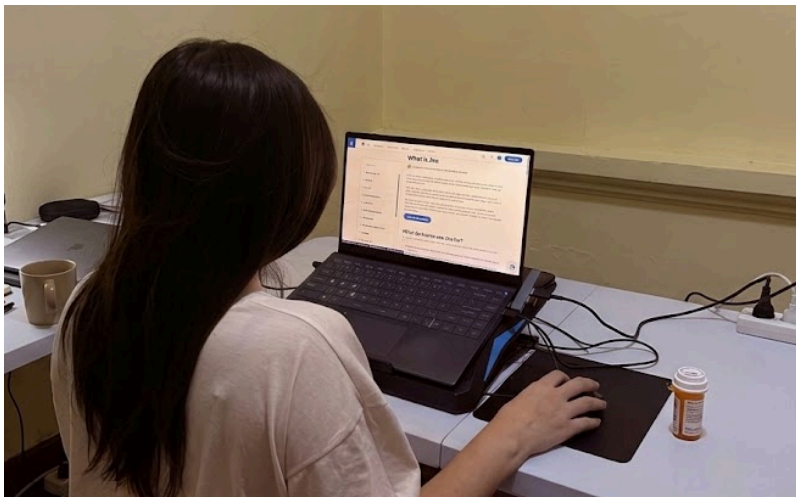


Figure 10. Defect Log and Progress Board

The figure shows Mary Pauleen C. Salvador, the Test Manager, watching a Jira tutorial.

XVI. Schedule

This level runs in the dynamic execution phase after MTP approval. Exact dates are set by the Test Manager once the instructor approves the MTP, and effort is estimated in person-hours. The testing is planned over two weeks, so the team can find and fix defects in the high-risk components before integration.

Period	Activity	Responsible	Effort (person-hours)

Week 1	Finalize test cases; set up framework, stubs, and data	Aduna / Marasigan	8
Week 1-2	Execute P1 component tests (PR-01, PR-02)	Aduna / Estolloso	10
Week 2	Execute P2/P3 tests (PR-08, PR-04, PR-07); retest	All testers	8

Table 3. Component Testing Schedule

XVII. Risks and Contingencies

Risks associated with component testing include source code that is not stable or not accessible, and a database schema that is missing a required constraint. To reduce these risks, the team agrees on a stable build and verifies the schema early. Contingency plans are ready in case a risk still occurs.

CR-01 – Source code not stable or not accessible for unit tests

Mitigation: Agree on a stable build with developers.

Contingency: Test through API endpoints with Postman instead.

CR-02 – Unique constraint not yet implemented in the schema

Mitigation: Verify the schema during the entry-criteria smoke check.

Contingency: Log a Critical defect and block PR-01 sign-off.

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A.Y. 2026-2027

In Partial Fulfillment of the Requirements in
IT 302 - System Integration and Architecture

APPENDIX B - PRELIMINARY REQUIREMENTS TRACEABILITY MATRIX (RTM)

of

Clinic Teleconsultation Booking and Reminder System

BY: *ADUNA, FRANZ JURIS U. || ESTOLLOSO, CIELO MARIE S. ||MARANAN, ARYANA RIZANNE N.*



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Submitted to:

Engr. Evangeline F. Gonzales

October 2026

APPENDIX B - PRELIMINARY REQUIREMENTS TRACEABILITY MATRIX (RTM)

I. Test Plan Identifier

ITP-IT302-MEDILYNK-1

II. References

Ref. No.	Document Title	Document Link
REF-MTP	MediLynk Master Test Plan v1.0	

Ref. No.	Document Title	Document Link
REF-06	MediLynk Activity 1 System Blueprint	MARANAN, Aryana Rizanne N._SIA_ACT 1 (1).pdf
REF-08	Possible Testing per Scenario and Recommended Database (Scenarios A-F: scope, testing ideas, data flow, main tables)	Possible Testing per Scenario and Recommended Database.pdf
REF-04	ISTQB Certified Tester - Foundation Level Syllabus v4.0	ISTQB Certified Tester - Foundation Level Syllabus v4.0

III. Introduction

Purpose

The purpose of this test plan is to verify that the MediLynk modules, the internal API layer, the database, and the third-party notification services communicate and exchange data correctly. The plan focuses on the Appointment-to-Notification and Appointment-to-Calendar flows. The integration testing phase is a crucial phase of development, as it ensures that the individual components work together as intended.

Scope

This test plan covers the following items:

- **TI-15:** inter-module flow.
- **TI-14:** the internal REST API.
- **TI-10:** Email, SMS, and Push integration.
- **TI-06:** the Notification Manager.
- **Calendar synchronization:** between the consultation/appointment service and the calendar service.

Third-party internals are excluded from this test plan.

Objective

The primary objective of this integration test is to confirm that the components of the system interact with one another without causing any problems and work as a whole. Specifically, the objectives of this test plan are the following:

1. To confirm that data flows correctly across module boundaries (REQ-03, REQ-04, REQ-06).
2. To confirm that a failed calendar sync is retried and flagged.
3. To confirm that a failed notification delivery is retried and flagged.

4. To find the P1 integration defects (PR-05) before system testing.

IV. Test Items

This test plan covers the following items of the MediLynk Teleconsultation System, identified from the Master Test Plan, the Activity 1 System Blueprint, and related documentation. Testing them ensures that what is delivered meets the specified requirements and maintains high-quality standards at the integration level.

1. **Inter-Module Integration / Data Flow (TI-15)**

   This item covers the data flow between modules: Appointment to Notification, Appointment to calendar, and Teleconsultation to Medical Records. It is linked to REQ-03 and REQ-04.

2. **Internal API Layer (TI-14)**

   This item covers the REST endpoints that connect the clients to the backend. It is linked to REQ-01 and REQ-03.

3. **Notification Manager (TI-06)**

   This item covers the Notification Manager, which uses the Observer pattern to trigger notifications. It is linked to REQ-04.

4. **Third-Party Service Integrations (TI-10)**

This item covers the Email, SMS, and Push integrations, tested through a sandbox or mock. It is linked to REQ-04.

5. **Appointment / Schedule Management Module (TI-02)**

At this level, this item covers the calendar synchronization. It is linked to REQ-03 and REQ-06.

V. Software Risk Issues

The product risks below apply to this test level. Scores follow the MTP (Likelihood x Impact), and full descriptions are in Appendix A.

- **Double booking of doctor and slot (PR-01, 20 / High):** A unique constraint on doctor + time slot, the conflict-check engine, and concurrency tests address this risk.
- **Unauthorized access to consultation history (PR-02, 20 / High):** Service-layer RBAC checks, a role x data decision table, and security scans address this risk.
- **Reminder timing failure (PR-03, 12 / Medium):** Validating the reminder calculation, BVA on the 24 h / 1 h lead times, and a provider-failure test (TCND-14) address this risk.
- **Calendar synchronization failure (PR-05, 16 / High):** Sync retry and flagging, together with state-transition and exploratory tests, address this risk.
- **Calendar not updated after reschedule/cancel (PR-08, 9 / Medium):** A calendar update on every reschedule/cancel and state consistency checks address this risk.

VI. Features To Be Tested

The following features will be tested in integration testing.

1. **Calendar Synchronization (REQ-03; TCND-04, 05)**
    - Availability updates within 5 seconds after booking, reschedule, and cancel.
    - A failed sync is queued, retried, and flagged.
2. **Reminders (REQ-04; TCND-06, 14)**
    - The Notification Manager triggers Email, SMS, and Push at 24 h and 1 h (tolerance 2 min) with correct details.
    - A provider failure is logged in Notification Logs, retried, and flagged without affecting the appointment.
3. **Status Change to Calendar (REQ-06; TCND-08)**
    - Reschedule and cancel trigger calendar updates through the API.
4. **Booking API (REQ-01; TCND-01, 02)**
    - Concurrent API requests yield one success.
    - Buffer violations are rejected.
5. **Authorization at API (REQ-02; TCND-03)**
    - The API rejects cross-role requests with the correct status code.

VII. Features Not To Be Tested

The following features will not be tested in this integration testing phase.

1. **Internal Operation of Third-Party Providers**

    These providers are the third party's responsibility.

2. **Full End-to-End User Scenarios**

    These scenarios are covered at the System level.

3. **Production-Scale Load**

This is excluded per MTP Section VII.

4. **Real Patient Data**

Data-privacy constraints rule out real patient data, so only synthetic data is used.

VIII. Approach

The integration testing approach is a key aspect of ensuring that the modules of MediLynk work together as intended. The team uses bottom-up integration, which starts with the highest-risk interface, the Appointment to Calendar flow (PR-05). It then moves on to the Notification and API layers. Sandbox or mock services replace the real third-party providers (TR-02), so that dispatch, failure, and retry behavior can be tested safely. The following sections describe the test tools and the integration flows used to evaluate the system.

Test Tools

The project team will use the following tools to support integration testing. Each tool is matched to a specific purpose.

Tool	Purpose
Postman	Automated API checks for booking, calendar, and notification endpoints.
JMeter	Small-scale concurrent booking and sync requests

Tool	Purpose
Notification sandbox / mock service	Verify dispatch, failure, and retry behavior
MySQL client	Verify Notification Logs and Appointments data
Jira / Trello	Defect tracking

Flowcharts

The integration flows under test are described step by step below (one flow per interface).

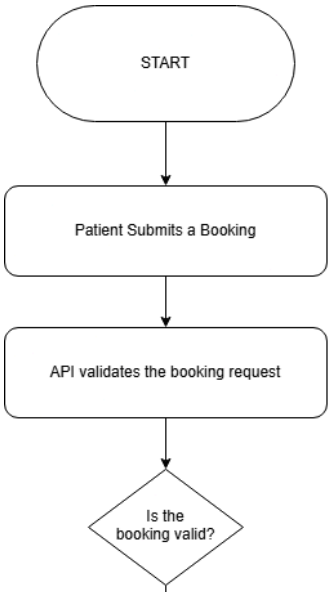


Figure 11. Booking to Calendar Integration Flowchart (REQ-03 and REQ-06)

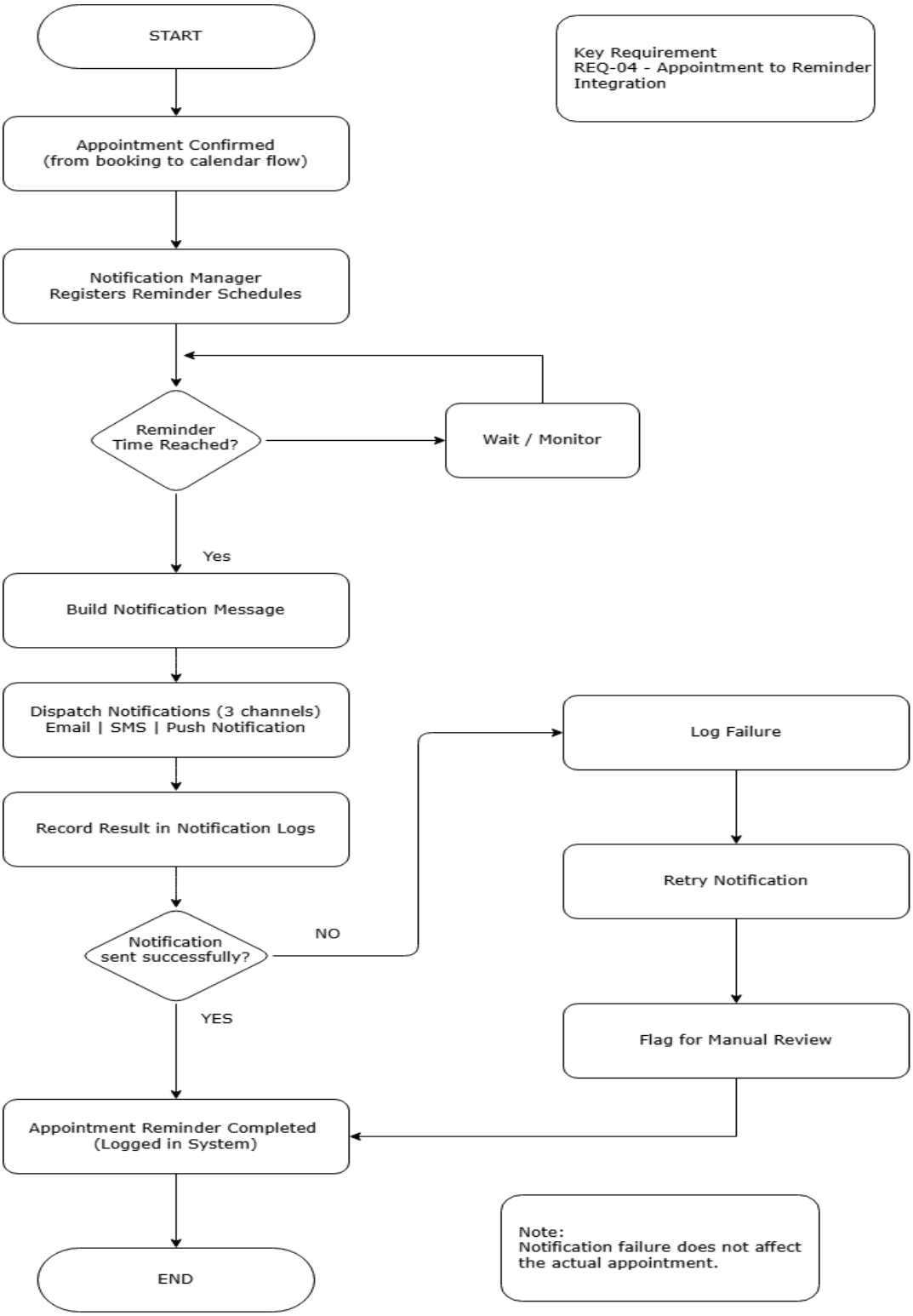


Figure 12. Appointment to Reminder Integration Flowchart (REQ-04)

IX. Item Pass/Fail Criteria

The following pass and fail criteria are designed to evaluate the integration-level behavior of the MediLynk Teleconsultation System under test, ensuring that it meets the specified requirements. The criteria below determine whether each test item is working as intended at the integration level.

Functionality	**Criteria**	
	Pass	**Fail**
Calendar Sync (REQ-03)	● Data is consistent across services within 5 seconds. ● A failure is retried and flagged.	● Data is stale. ● A failure is not retried or flagged.
Reminder Flow (REQ-04)	● Reminders are sent within tolerance. ● Reminders carry the correct data and channels.	● A reminder is late, wrong, or missing. ● A reminder is unlogged.
Status to Calendar (REQ-06)	● The calendar reflects each transition.	● The calendar is outdated or conflicting.

Functionality	Criteria	
	Pass	**Fail**
Booking API (REQ-01)	● One booking succeeds for concurrent requests.	● A double booking is accepted.
API Authorization (REQ-02)	● Unauthorized calls are denied.	● Unauthorized data is returned.

X. Suspension Criteria and Resumption Requirements

This section declares the suspension criteria and resumption requirements for the integration testing of MediLynk. Testing can be paused for the time needed for recovery when any of the following occurs:

1. **Service Unavailable:** The calendar service or the notification mock/sandbox is unavailable.
2. **Blocking Defect in a High-Priority Risk:** A blocking defect in PR-01, PR-02, or PR-05 halts dependent tests.
3. **Smoke Check Failure:** Component-level smoke checks fail on the build.

Resumption Requirements:

Testing activities will only recommence once the following requirements are met:

1. **Blocking Defect Resolved:** The blocking defect is fixed and re-verified.

2. **Environment Restored:** The environment or service is restored and confirmed stable.
3. **Updated Build and Test Data:** The updated build and test data are delivered and smoke-tested.
4. **Test Manager Approval:** The Test Manager approves resumption.
5. **Mitigation Confirmed:** The relevant Appendix A mitigation is confirmed.

XI. Test Deliverables

The following deliverables are produced during the integration testing of MediLynk:

1. **Integration Test Cases and Postman Collections:** The test cases and the Postman collections used to verify each interface.
2. **Interface Test Logs and Screenshots:** The execution results and the screenshots that serve as evidence.
3. **Defect Reports and Integration Status Report:** The defects found during integration testing and the report on testing progress.

XII. Testing Task

The following testing tasks must be completed during integration testing:

1. **Finalize Integration Test Cases:** Finalize the integration test cases (TCND-01 to 06, 08, and 14). *Owner: Test Analyst.*
2. **Configure Test Services:** Configure the sandbox/mock notification services and the calendar service. *Owner: Technical Test Analyst.*
3. **Prepare Test Data:** Prepare the date-sensitive appointment and reminder data. *Owner: Integration Test Engineer.*

4. **Execute Tests:** Execute Flow 1, then Flow 2, then the API checks. *Owner: Integration Test Engineer / All testers.*
5. **Log and Retest Defects:** Log the defects found and retest them after the fixes. *Owner: Test Analyst.*
6. **Regression and Reporting:** Run regression testing after the fixes and report the results. *Owner: Test Manager.*

XIII. Environmental Needs

1. **Workstations:** Laptop or desktop with an Intel Core i5 or equivalent, 8 GB RAM, and 256 GB of free storage, running Windows 10/11.
2. **Browsers / Mobile:** Google Chrome and Microsoft Edge, plus an Android device or emulator. iOS is not available.
3. **Test Data:** Synthetic or anonymized data only, with accounts for Patient, Doctor, Clinic Admin/Staff, and System Administrator.
4. **Network:** A stable internet connection for the sandboxes, or a local network if the system is hosted locally.
5. **Services:** Reachable REST endpoints (TI-14), the calendar service, and the provider sandboxes or mocks (TI-10).
6. **Tools:** Postman, JMeter, and the MySQL client.

XIV. Responsibilities

In order to guarantee that integration testing is carried out successfully, each member of the team is assigned a role with specific duties at this level.

- **Integration Test Engineer – Estolloso, Cielo Marie S.:** Leads the integration testing and the interface data flow.
- **Test Analyst – Aduna, Franz Juris U.:** Designs the integration test cases.
- **Technical Test Analyst – Marasigan, Kylie Denise D.G.:** Handles the environment and tooling.
- **Test Manager – Salvador, Mary Pauleen C.:** Monitors and controls the testing.

XV. Staffing and Training Needs

Integration testing needs team members who have knowledge of REST APIs, JSON, and the Observer pattern, so that the interfaces between the MediLynk modules can be tested properly. The members assigned to each role are listed in Section XIV.

1. **Training for Testing Team:** Training refers to the learning opportunities given to the testing team to build the skills needed for integration testing. The following training will be provided:
    - **API and Integration Orientation:** Members learn how the REST endpoints and the modules connect and exchange data.
    - **Database Testing Orientation:** Members learn how to check the stored data, such as Notification Logs and Appointments, directly in the database.

- **Sandbox Setup:** Members learn how to set up the sandbox or mock services that replace the real providers.

XVI. Schedule

This level runs in the dynamic execution phase after MTP approval. Exact dates are set by the Test Manager once the instructor approves the MTP, and effort is estimated in person-hours. The testing is planned over two weeks, starting after component testing, so the team can verify the interfaces between the modules before system testing.

Testing activity	Week 1	Week 2	Week 3
Set up services; finalize cases			
Flow 1 (PR-05) and API checks (PR-01)			
Flow 2 (PR-03), provider failure, regression			

Table 4. Integration Testing Schedule

XVII. Risks and Contingencies

Risks associated with integration testing include an unavailable sandbox, a calendar service that is not ready, and test data that expires. To reduce these risks, the team uses a mock notification service, confirms service availability at entry, and regenerates the data after each fix cycle. Contingency plans are ready in case a risk still occurs.

IR-01 – Sandbox unavailable or quota exhausted (TR-02)

Mitigation: Use a mock notification service.

Contingency: Rely on the mock and record real delivery as a residual risk.

IR-02 – Calendar service not ready

Mitigation: Confirm service availability at entry.

Contingency: Use a stub calendar service and flag it as a limitation.

IR-03 – Date-sensitive test data expires

Mitigation: Regenerate data after each fix cycle.

C*ontingency:* Recreate the data manually.

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A.Y. 2026-2027

In Partial Fulfillment of the Requirements in
IT 302 - System Integration and Architecture

APPENDIX C - EARLY TESTING / STATIC REVIEW LOG

of

Clinic Teleconsultation Booking and Reminder System

BY: *MARANAN, ARYANA RIZANNE N.. || SALVADOR, MARY PAULEEN C. || TAMAYO, JONIE R.*

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Submitted to:

Engr. Evangeline F. Gonzales

October 2026

APPENDIX C - EARLY TESTING / STATIC REVIEW LOG

I. Test Plan Identifier

STP-IT302-MEDILYNK-1

II. References

Ref. No.	Document Title	Document Link
REF-MTP	MediLynk Master Test Plan v1.0	
REF-01	Assigned Scenario D	IT302_Midterm_Project_Master_Test_Plan_Studen.pdf
REF-06	MediLynk Activity 1 System Blueprint	MARANAN, Aryana Rizanne N._SIA_ACT 1 (1).pdf
REF-04	ISTQB CTFL Syllabus v4.0.1 (system testing, test quadrants)	IEEE SA - IEEE 829-2008

III. Introduction

Purpose

The purpose of this test plan is to verify that the complete, integrated MediLynk system meets all ten requirements end to end through the web and mobile clients. The verification includes security, recovery, performance under normal conditions, and compatibility. The system testing phase is a crucial phase of development, as it checks the system as a whole after the individual components and interfaces have been tested.

Scope

This test plan covers REQ-01 to REQ-10 on the full system, which includes the following:

- **Web clients:** Chrome and Edge.
- **Mobile client:** Android.
- **Backend:** the Express backend.
- **Database:** the MySQL database.
- **Notification services:** sandbox/mock notification services.
- **Video session:** the video simulator.

The following are excluded from this test plan:

- Production-scale load.
- iOS.
- Third-party internals.

Objective

The primary objective of this system test is to confirm that the integrated system works as a whole and meets its requirements. Specifically, the objectives of this test plan are the following:

1. To confirm that the system satisfies every pass criterion in MTP Section IX.
2. To confirm that the P1 risks show no unresolved Critical defect.
3. To confirm that recovery, masking, and video-session behavior meet the agreed thresholds (ER-06 to ER-08).

IV. Test Items

This test plan covers the following items of the MediLynk Teleconsultation System, identified from the Master Test Plan, the Activity 1 System Blueprint, and related documentation. Testing them ensures that what is delivered meets the specified requirements and maintains high-quality standards at the system level.

1. **All In-Scope Modules (TI-01 to TI-15)**

These are all in-scope modules, the database, the APIs, the integrations, and the screens, tested as one system. They are linked from REQ-01 to REQ-10.

2. **Teleconsultation Management Module (TI-04)**

This module manages the video and call sessions. It is linked to REQ-08.

3. **Service Recovery / Fallback Handler (TI-12)**

This item covers the recovery logic for interrupted teleconsultation sessions. It is linked to REQ-07.

4. **Medical Records Masking and RBAC (TI-07 / TI-11)**

This item covers the masking of medical records and the role-based access control. It is linked to REQ-05 and REQ-02.

5. **User Interface / Screens (TI-13)**

This item covers the screens of the web and mobile clients. It is linked to REQ-09.

V. Software Risk Issues

The product risks below apply to this test level. Scores follow the MTP (Likelihood x Impact), and full descriptions are in Appendix A.

- **Double booking of doctor and slot (PR-01, 20 / High):** A unique constraint on doctor + time slot, the conflict-check engine, and concurrency tests address this risk.
- **Unauthorized access to consultation history (PR-02, 20 / High):** Service-layer RBAC checks, a role x data decision table, and security scans address this risk.
- **Calendar synchronization failure (PR-05, 16 / High):** Sync retry and flagging, together with state-transition and exploratory tests, address this risk.
- **Missing data masking (PR-06, 15 / Medium):** Masking of sensitive fields, a role x field matrix, and a privacy review address this risk.
- **Reminder timing failure (PR-03, 12 / Medium):** Validating the reminder calculation, BVA on the 24 h / 1 h lead times, and a provider-failure test (TCND-14) address this risk.

- **Video platform unavailable or call drops (PR-09, 12 / Medium):** A documented recovery option (Rejoin, then a reschedule prompt) and a simulated outage address this risk.

VI. Features To Be Tested

The following features will be tested in system testing.

1. **Booking and Expired Slot (REQ-01, 10; TCND-01, 02, 12)**
    - End-to-end booking.
    - Simultaneous requests.
    - Buffer edge.
    - Past-date rejection.
2. **Access and Masking (REQ-02, 05; TCND-03, 07)**
    - Role x data and role x field matrices for all four roles.
    - OWASP ZAP scan.
3. **Sync and Reminders (REQ-03, 04; TCND-04, 05, 06, 13)**
    - The full journey: register, select doctor, book, calendar updates, and the 24 h and 1 h reminders.
4. **Recovery and Video (REQ-07, 08; TCND-09, 10)**
    - An outage or dropped call shows the Rejoin option, then a reschedule prompt, to both parties.
    - The session starts within 10 seconds and lasts for 10 minutes.
5. **Registration (REQ-09; TCND-11)**
    - UI and server rejection of empty, invalid, and boundary data.

6. **Compatibility (Chrome, Edge, Android)**
    - Core flows work on all supported clients.

VII. Features Not To Be Tested

The following features will not be tested in this system testing phase.

1. **Production-Scale Load and Stress Testing**

    Free-tier quotas apply, and no performance risk is above Medium.

2. **Selenium/Playwright Automation**

    This is planned for later, in the final project.

3. **iOS and Wider Browser Matrix**

    These are not available.

4. **Reporting, Symptom & File, and Export Modules**

    These modules are excluded per MTP Section VII.

5. **Full UAT**

    Full UAT is covered by the Acceptance Test Plan.

VIII. Approach

The system testing approach is manual end-to-end testing with targeted automation. Testing is executed in risk order: P1 (PR-01, PR-02, PR-05) first, then P2 starting with PR-06, then P3. The test quadrants are weighted as follows: Q1 and Q4 for concurrency, security, and sync; Q2 for business rules; and Q3 for exploratory sessions and a usability walkthrough. The following sections describe the test tools and techniques used to evaluate the system.

Test Tools

The project team will use the following tools to support system testing. Each tool is matched to a specific purpose.

Tool	Purpose
Postman	API-level checks during system scenarios
JMeter	Small-scale concurrent booking; video-start timing
OWASP ZAP	Security and authorization scans
Video simulator	Outage and dropped-call simulation
Jira / Trello	Defect tracking

Test Techniques

The team applies the following techniques to design the test cases:

- **Decision Table Testing:** Applied to the roles.
- **State Transition Testing:** Applied to the status and sync.
- **Boundary Value Analysis (BVA):** Applied to the dates and lead times.
- **Error Guessing:** Applied to the outage and concurrency scenarios.
- **Exploratory Testing:** Applied to the timing and the video session.
- **Checklist-Based Testing:** Applied to the video session and the smoke checks.

IX. Item Pass/Fail Criteria

The following pass and fail criteria are designed to evaluate the system-level behavior of the MediLynk Teleconsultation System under test, ensuring that it meets the specified requirements. The criteria below determine whether each test item is working as intended at the system level.

	Criteria	
Functionality	**Pass**	**Fail**
Booking (REQ-01, 10)	● Only one booking is made per slot. • Buffer and expired slots are rejected.	● A double booking is accepted. • An expired slot is accepted.

Access and Masking (REQ-02, 05)	● No data is visible outside role clearance.	● Any unauthorized visibility occurs.
Sync and Reminders (REQ-03, 04)	● Updates are within 5 seconds and reminders within the 2-minute tolerance. • The data is correct.	● A threshold is breached. • The data is missing.
Recovery and Video (REQ-07, 08)	● The recovery option is shown to both parties. • The session starts within 10 seconds.	● No option is shown. • The option is shown to only one party. • A threshold is breached.
Registration (REQ-09)	● Invalid data is rejected with a clear error.	● Invalid data is accepted.

X. Suspension Criteria and Resumption Requirements

This section declares the suspension criteria and resumption requirements for the system testing of MediLynk. Testing can be paused for the time needed for recovery when any of the following occurs:

1. **Smoke Check Failure:** Critical smoke checks fail on the build.
2. **Blocking P1 Defect:** A P1 blocking defect prevents dependent scenarios.
3. **Services Unavailable:** The mock/sandbox services or the video simulator are unavailable.
4. **Environment or Data Unavailable:** The test environment or test data is unavailable.

Resumption Requirements:

Testing activities will only recommence once the following requirements are met:

1. **Blocking Defect Resolved:** The blocking defect is fixed and re-verified.
2. **Environment Restored:** The environment or service is restored and confirmed stable.
3. **Updated Build and Test Data:** The updated build and test data are delivered and smoke-tested.
4. **Test Manager Approval:** The Test Manager approves resumption.
5. **Mitigation Confirmed:** The relevant Appendix A mitigation is confirmed.

XI. Test Deliverables

The following deliverables are produced during the system testing of MediLynk:

1. **System Test Cases and Checklists:** The test cases and checklists used to verify the system.
2. **Test Logs, Screenshots, and Evidence:** The execution results and the evidence that supports them.
3. **Defect Reports, Progress Reports, and Test Completion Report Inputs:** The defects found, the progress of testing, and the inputs for the Test Completion Report.

XII. Testing Task

The following testing tasks must be completed during system testing:

1. **Finalize System Test Cases and Checklists:** *Owner: Test Analyst.*
2. **Configure the Full Environment:** Configure the full environment, ZAP, JMeter, and the video simulator. *Owner: Technical Test Analyst.*
3. **Load Synthetic Data:** Load the synthetic data for all four roles. *Owner: Integration Test Engineer.*
4. **Execute Scenarios:** Execute the P1 scenarios, then P2, then P3. *Owner: All testers.*
5. **Exploratory Sessions and Usability Walkthrough:** *Owner: Integration Test Engineer / QA Engineer.*
6. **Retest, Regression, and Test Completion Report:** *Owner: Test Manager.*

XIII. Environmental Needs

1. **Workstations:** Laptop or desktop with an Intel Core i5 or equivalent, 8 GB RAM, and 256 GB of free storage, running Windows 10/11.
2. **Browsers / Mobile:** Google Chrome and Microsoft Edge, plus an Android device or emulator. iOS is not available.
3. **Test Data:** Synthetic or anonymized data only, with accounts for Patient, Doctor, Clinic Admin/Staff, and System Administrator.
4. **Application Server:** The Node.js/Express build.
5. **Database:** MySQL (InnoDB) with the six tables.
6. **Video and Notifications:** A mock video simulator, and an Email/SMS/Push sandbox or mock.
7. **Tools:** Postman, JMeter, OWASP ZAP, and Jira/Trello.

XIV. Responsibilities

In order to guarantee that system testing is carried out successfully, each member of the team is assigned a role with specific duties at this level.

Technical Test Analyst – Marasigan, Kylie Denise D.G.: Handles the security, performance, compatibility, and recovery tests.

Test Analyst – Aduna, Franz Juris U.: Handles the functional scenarios and expected results.

QA Engineer – Tamayo, Jonie R.: Handles the quality review and the usability walkthrough.

Integration Test Engineer – Estolloso, Cielo Marie S.: Handles the data preparation and the end-to-end and exploratory runs.

Test Manager – Salvador, Mary Pauleen C.: Handles the monitoring and reporting.

XV. Staffing and Training Needs

System testing needs team members with functional, security, and exploratory testing skills and tool knowledge, so that the whole of MediLynk can be tested end to end. The members assigned to each role are listed in Section XIV.

1. **Training for Testing Team:** Training refers to the learning opportunities given to the testing team to build the skills needed for system testing. The following training will be provided:
    - **Security and Privacy Orientation:** Members learn how to check access control, data masking, and privacy requirements.
    - **Defect Reporting Training:** Members learn how to write clear and complete defect reports.
    - **Environment Setup:** Members learn how to set up the full test environment and tools.

XVI. Schedule

This level runs in the dynamic execution phase after MTP approval. Exact dates are set by the Test Manager once the instructor approves the MTP, and effort is estimated in person-hours. The testing is planned over two weeks, starting after integration testing begins, so the team can verify the complete system before acceptance.

Period	Activity	Responsible	Effort (person-hours)
Week 3	Environment ready; smoke check; P1 scenarios	Marasigan / Aduna	12
Week 3-4	P2 scenarios; exploratory and usability sessions	All testers	14
Week 4	P3 scenarios; retest; regression; report	All testers	10

Table 5. System Testing Schedule

XVII. Risks and Contingencies

Risks associated with system testing include an open role x field matrix, thresholds that are still pending confirmation, and a video simulator that differs from the real platform. To reduce these risks, the team completes the matrix before execution, seeks confirmation early, and documents the simulator's limits. Contingency plans are ready in case a risk still occurs.

SR-01 – Role x field matrix still open (ER-08)

Mitigation: Complete the matrix before execution.

Contingency: Test with the team-defined matrix and record it as a residual risk.

SR-02 – Thresholds pending instructor confirmation (ER-06 to ER-08)

Mitigation: Seek confirmation early.

Contingency: Use the team-defined values and record them as a residual risk.

SR-03 – Video simulator differs from the real platform

Mitigation: Document the simulator's limits.

Contingency: Record the real-platform behavior as unverified.

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A.Y. 2026-2027

In Partial Fulfillment of the Requirements in
IT 302 - System Integration and Architecture

APPENDIX D - TESTING QUADRANTS COVERAGE MAP

of

Clinic Teleconsultation Booking and Reminder System

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October 2026

APPENDIX D - TESTING QUADRANTS COVERAGE MAP

I. Test Plan Identifier

ATP-IT302-MEDILYNK-1

II. References

Ref. No.	Document Title	Document Link
REF-MTP	MediLynk Master Test Plan v1.0 (Appendix D, Q3; Appendix E, Principle 7)	
REF-01	Assigned Scenario D	IT302_Midterm_Project_Master_Test_Plan_Studen.pdf
REF-06	MediLynk Activity 1 System Blueprint	MARANAN, Aryana Rizanne N._SIA_ACT 1 (1).pdf
REF-04	ISTQB CTFL Syllabus v4.0.1 (acceptance testing)	ISTQB Certified Tester - Foundation Level Syllabus v4.0

III. Introduction

Purpose

To plan the validation that MediLynk meets business needs and is acceptable to its intended users - Patients, Doctors, and Clinic Admin/Staff - guarding against the absence-of-defects fallacy (Principle 7).

Scope

At Midterm this is planning only; full UAT execution is not required. The plan covers acceptance scenarios for booking, reminders, recovery, video consultation, and registration, using synthetic data and the same environment as system testing.

Objective

To define acceptance scenarios, participants, entry/exit criteria, and pass/fail criteria so that user acceptance can be executed after system testing, and to confirm the system is fit for its intended clinic use.

IV. Test Items

This test plan covers the following items of the MediLynk Teleconsultation System, identified from the Master Test Plan, the Activity 1 System Blueprint, and related documentation. Testing them ensures that what is delivered meets the specified requirements and maintains high-quality standards at the acceptance level.

ID	Test Item	Linked REQ / Notes
UAT-1	Patient journey: register, book, receive reminders, join consultation	REQ-01, 04, 08, 09
UAT-2	Doctor journey: view schedule and assigned consultations, conduct session, recovery	REQ-02, 03, 07, 08
UAT-3	Clinic Admin/Staff journey: manage appointments, cancel expired booking, review flagged sync	REQ-03, 06, 10
UAT-4	Privacy check: sensitive data hidden from roles without clearance	REQ-05

V. Software Risk Issues

The product risks below apply to this test level. Scores follow the MTP (Likelihood x Impact); full descriptions are in Appendix A.

ID	Risk	Score / Level	Mitigation / Test Response
PR-01	Double booking of doctor and slot	20 / High	Unique constraint on doctor + time slot; conflict-check engine; concurrency tests
PR-03	Reminder timing failure	12 / Medium	Validate reminder calculation; BVA on 24 h / 1 h lead times; provider-failure test (TCND-14)
PR-06	Missing data masking	15 / Medium	Mask sensitive fields; role x field matrix; privacy review
PR-09	Video platform unavailable or call drops	12 / Medium	Documented recovery option (Rejoin, then reschedule prompt); simulated outage

VI. Features To Be Tested

Requirement	Feature	What Will Be Verified at this Level
REQ-01, 04, 09	Patient booking and reminders (TCND-13)	A Patient completes registration and booking easily and receives 24 h and 1 h reminders.
REQ-07, 08	Video and recovery (TCND-09, 10)	Users can start a session and follow the Rejoin then reschedule prompt when a call drops.
REQ-03, 06	Doctor and staff calendar (TCND-04, 08)	Doctors and staff see accurate schedules after changes.

Requirement	Feature	What Will Be Verified at this Level
REQ-05	Data masking (TCND-07)	Business users confirm sensitive information is properly hidden.

VII. Features Not To Be Tested

Feature / Area	Reason for Exclusion
Full UAT execution at Midterm	Not required; planning only.
Real patient data	Data-privacy constraints.
iOS and production hosting	Not available.
Reporting Module	Excluded per MTP Section VII.

VIII. Approach

Scenario-based, business-facing acceptance (Q2/Q3) run by representative users in the role they play. The Q3 usability walkthrough of booking and recovery screens supports this plan.

Test Tools

Tool	Purpose
Acceptance scenario checklists	Scripted user steps with expected outcomes
Jira / Trello	Record feedback and defects
Screen capture	Evidence of results

Entry criteria: system test exit criteria met; no open Critical defect on P1 features; UAT participants available.

Exit criteria: all acceptance scenarios run; every business-critical scenario (UAT-1, UAT-2) passes; open issues listed with impact; sign-off recorded by the instructor.

IX. Item Pass/Fail Criteria

Test Item	Pass Criteria	Fail Criteria
UAT-1 Patient journey	Completed without assistance; correct reminders received.	Journey blocked or wrong reminder.
UAT-2 Doctor journey	Assigned data only; recovery prompt clear.	Wrong data shown or recovery unclear.
UAT-3 Staff journey	Calendar accurate; expired booking cancellable.	Calendar wrong or action impossible.
UAT-4 Privacy	Sensitive fields hidden as agreed.	Sensitive data visible.

X. Suspension Criteria and Resumption Requirements

Suspension criteria:

- System test exit criteria not met.
- Critical defect discovered during UAT.
- UAT environment or participants unavailable.

Resumption requirements:

- Blocking defect fixed and re-verified; environment or service restored and confirmed stable.
- Updated build and test data delivered and smoke-tested.
- Test Manager approves resumption, and the relevant Appendix A mitigation is confirmed.

XI. Test Deliverables

- Acceptance test scenarios and checklists
- UAT results and feedback log
- Sign-off record and residual-risk statement

XII. Testing Task

No.	Testing Task	Owner (role)
1	Finalize acceptance scenarios from TCND-13, 09, 10, 04, 07	QA Engineer
2	Select participants and prepare synthetic accounts	Integration Test Engineer
3	Brief participants; run UAT sessions	QA Engineer / Test Analyst
4	Log feedback and defects	All testers
5	Report results and obtain sign-off	Test Manager

XIII. Environmental Needs

Requirement	Specification
Workstations	Laptop/desktop, Intel Core i5 or equivalent, 8 GB RAM, 256 GB free storage; Windows 10/11
Browsers / Mobile	Google Chrome, Microsoft Edge; Android device or emulator (iOS not available)
Test data	Synthetic or anonymized data only; accounts for Patient, Doctor, Clinic Admin/Staff, and System Administrator
Environment	Stable build matching the system-tested version with sandbox/mock services

XIV. Responsibilities

Role	Assigned Member	Responsibility at this Level
QA Engineer	Tamayo, Jonie R.	Prepare scenarios; run and evaluate UAT sessions
Integration Test Engineer	Estolloso, Cielo Marie S.	Accounts, data, and session logistics
Test Analyst	Aduna, Franz Juris U.	Scenario design support
Test Manager	Salvador, Mary Pauleen C.	Report and sign-off coordination
Instructor	Engr. Evangeline F. Gonzales	Acceptance authority

XV. Staffing and Training Needs

UAT participants need only a short orientation; facilitators need scenario-writing and feedback-recording skills.

- Training: brief participants on scenarios, test accounts, and responsible handling of test data.

XVI. Schedule

This level runs in the dynamic execution phase after MTP approval. Exact dates are set by the Test Manager once the instructor approves the MTP; effort is estimated in person-hours.

Period	Activity	Responsible	Effort (person-hours)
Week 4-5	Prepare scenarios and participants	Tamayo / Estolloso	6
Week 5	Run UAT sessions	Tamayo / Aduna	8
Week 5	Report and sign-off	Salvador	4

XVII. Risks and Contingencies

ID	Risk	Mitigation	Contingency
AR-01	Participants unavailable	Schedule early	Use team members in role
AR-02	Late system defects delay UAT	Enforce system exit criteria	Run UAT on stable subset

BULACAN STATE UNIVERSITY

College of Information and Communication Technology

Pimentel Hall, City of Malolos

A.Y. 2026-2027

In Partial Fulfillment of the Requirements in
IT 302 - System Integration and Architecture

APPENDIX E: SEVEN TESTING PRINCIPLES EVIDENCE MATRIX

of

Clinic Teleconsultation Booking and Reminder System

BY: *ESTOLLOSO, CIELO MARIE S.* || *MARASIGAN, KYLIE DENISE D.G.* || *TAMAYO, JONIE R.*

MediLynk Solutions - BSIT 3IG2

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ESTOLLOSO, Cielo Marie S.

MARANAN, Aryana Rizanne N.

MARASIGAN, Kylie Denise D.G.

SALVADOR, Mary Pauleen C.

TAMAYO, Jonie R.

Submitted to:

Engr. Evangeline F. Gonzales

October 2026

APPENDIX E: SEVEN TESTING PRINCIPLES EVIDENCE MATRIX

I. Test Plan Identifier

APX-E-IT302-MEDILYNK-1

II. References

Ref. No.	Document Title	Document Link
REF-MTP	MediLynk Master Test Plan v1.0 (Sections I to XIX; Appendices A to D; Component, Integration, System, and Acceptance Test Plans)	
REF-02	IT302 Midterm Project Student Guide (Section 4 Core Rule; Appendix E template; Section 14 rubric, criterion 8)	IT302_Midterm_Project_Master_Test_Plan_Studen
REF-04	ISTQB CTFL Syllabus v4.0.1 (Sec. 1.3 Testing Principles)	ISTQB Certified Tester - Foundation Level Syllabus v4.0
REF-06	MediLynk Activity 1 System Blueprint (architecture and feature list)	MARANAN, Aryana Rizanne N._SIA_ACT 1 (1).pd

III. Introduction

Purpose

To plan the validation that MediLynk meets business needs and is acceptable to its intended users: Patients, Doctors, and Clinic Admin/Staff. The plan guards against the absence-of-defects fallacy (ISTQB Principle 7): a system can pass every system test and still fail to help the people who must use it. Acceptance testing therefore asks whether the system is fit for use in a clinic, not only whether it matches its specification.

Scope

At Midterm this is planning only; full UAT execution is not required. The plan covers acceptance scenarios for booking, reminders, recovery, video consultation, and registration, using synthetic data and the same environment as system testing. It defines who will take part, what each participant will do, and how a result will be judged, so that UAT can be executed after system testing without further design work.

Objective

To define acceptance scenarios, participants, entry/exit criteria, and pass/fail criteria so that user acceptance can be executed after system testing, and to confirm the system is fit for its intended clinic use. More specifically:

- To confirm that each user role can finish its main journey without assistance.
- To confirm that business users find sensitive information properly hidden.
- To collect usability feedback on booking and recovery screens.

- To give the instructor a clear basis for sign-off, including a list of open issues and residual risks.

IV. Test Items

This test plan covers the following items of the MediLynk Teleconsultation System, identified from the Master Test Plan, the Activity 1 System Blueprint, and related documentation. Testing them ensures that what is delivered meets the specified requirements and maintains high-quality standards at the acceptance level.

ID	Test Item	Linked REQ / Notes
UAT-1	Patient journey: register, book, receive reminders, join consultation	REQ-01, 04, 08, 09
UAT-2	Doctor journey: view schedule and assigned consultations, conduct session, recovery	REQ-02, 03, 07, 08
UAT-3	Clinic Admin/Staff journey: manage appointments, cancel expired booking, review flagged sync	REQ-03, 06, 10
UAT-4	Privacy check: sensitive data hidden from roles without clearance	REQ-05

Acceptance Scenario Outline

Scenario	Role	Steps (summary)	Expected Outcome
UAT-1	Patient	Register; pick a doctor and slot; confirm booking; receive 24 h and 1 h reminders; join the video session; handle a dropped call using Rejoin.	Journey is completed unaided; reminders are correct; recovery is clear.
UAT-2	Doctor	Open the schedule; view only assigned consultations; start a session; experience a dropped call and follow the recovery prompt.	Only assigned data is shown; recovery prompt is clear.
UAT-3	Clinic Admin/Staff	Create, change, and cancel appointments; cancel an expired booking; review a flagged sync item.	Calendar is accurate; every action is possible.
UAT-4	Staff / Doctor (no clearance)	Open records and look for sensitive fields.	Sensitive fields are hidden as agreed.

V. Software Risk Issues

The product risks below apply to this test level. Scores follow the MTP (Likelihood x Impact); full descriptions are in Appendix A.

ID	Risk	Score / Level	Mitigation / Test Response
PR-01	Double booking of doctor and slot	20 / High	Unique constraint on doctor + time slot; conflict-check engine; concurrency tests. In UAT, users confirm that a taken slot is clearly shown as unavailable.
PR-03	Reminder timing failure	12 / Medium	Validate reminder calculation; BVA on 24 h / 1 h lead times; provider-failure test (TCND-14). In UAT, patients confirm the reminders they receive are correct and on time.
PR-06	Missing data masking	15 / Medium	Mask sensitive fields; role x field matrix; privacy review. In UAT, business users confirm masking.

PR-09	Video platform unavailable or call drops	12 / Medium	Documented recovery option (Rejoin, then reschedule prompt); simulated outage. In UAT, users judge whether the prompt is understandable.

VI. Features To Be Tested

Each row is one evidence check. The Required Evidence line comes from the Student Guide template. The last column gives the places in the MTP that the reviewer will verify.

#	Testing Principle	MediLynk-Specific Evidence	Where the Group Shows It
1	Testing shows presence, not absence of defects *Required: no "defect-free" claim; realistic exit criteria and residual-risk thinking.*	• Exit criteria never require zero defects. They require all P1 test conditions executed, no unresolved Critical defect unless the Test Manager records formal risk acceptance, and at least 80% of P2/P3 conditions executed, with any unexecuted condition listed with its reason and residual risk.	Section IX: Exit Criteria Section III: Assumptions Section XVII: TR-02 Section XI: Test Completion Report Section XIX: Residual Risk

		• The plan says that meeting the exit criteria shows the planned testing was sufficient, not that the system is defect-free. • Named residual risks: the open role x field matrix and unconfirmed thresholds (ER-06 to ER-08), and real third-party delivery that stays unverified when the mock is used (TR-02).	
2	Exhaustive testing is impossible *Required: scope, prioritization, techniques, and coverage choices are explicit.*	• Scope has ten requirements (REQ-01 to REQ-10) and nine product risks (PR-01 to PR-09). Three of the sixteen test items (TI-03, TI-08, TI-16) are excluded with reasons, along with production-scale load, iOS, Selenium, and real patient data. • Techniques replace “test every input”: BVA on the 10-minute buffer edge, expired and boundary dates, and the 24 h / 1 h reminder	Section III: Scope Section IV: Test Items Section V: Risk Issues Section VII: Not To Be Tested Section VIII: Techniques Appendices A and B

		lead times; EP on registration fields; a decision table on role x data combinations. • Risk scores set the P1/P2/P3 order. Coverage goals are 10 of 10 REQs, 9 of 9 PRs, all P1 conditions, and at least 80% of P2/P3.	
3	Early testing saves time and money *Required: static review and risk analysis completed before detailed execution; findings corrected.*	• Static review was done at MTP v0.1 (September 16, 2026). Risk analysis followed on September 17, before the test approach (September 22) and before any test design or execution. • Findings were corrected before baseline: v1.0 added ER-06 to ER-08 (including the 10-minute buffer rule and the role x field matrix) and PR-09, plus exit criteria, a severity scale, and monitoring. • A peer static review by Tamayo, Jonie R. (QA Engineer) was incorporated into v1.0.	Appendix C: Static Review Log Appendix A: Risk Matrix Section I: Revision History Section XVI: Schedule (Days 2, 3, 10)

4	Defects cluster together *Required: high-risk or complex areas receive deeper or earlier test attention.*	• PR-01 (20), PR-02 (20), and PR-05 (16) are High and are tested first, at several levels and with several techniques. • TI-02 Appointment / Schedule Management is the most connected module (REQ-01, 03, 06, 10), so it carries component, integration, and system coverage. • Monitoring counts defects by severity and by test item. A cluster in one item triggers extra tests or an exploratory session there.	Section V: Summary of product risks Section IV: TI-02 Section VIII: Test Sequence; Monitoring and Control Appendices A and B (P1 rows)
5	Tests wear out *Required: plan includes review and refresh of regression tests and test data after changes or repetition.*	• Remaining Test Task 7 plans regression after every fix. Task 8 reviews test cases and data for outdated or ineffective material and refreshes them after each fix cycle. • Date-sensitive appointment and reminder data is regenerated after each fix cycle, because expired slots	Section XII: Tasks 7 and 8 Section IX: Exit Criteria Integration Test Plan: IR-03 Section XIX: Regression Testing

		and past reminder times make old data useless (IR-03). • Fixed P1 defects must be retested and the planned regression set completed before exit.	
6	Testing is context dependent *Required: test approach and quadrant mix are tailored to the assigned scenario.*	• The approach follows Scenario D: schedule overlap, double booking, reminders, confidential health data, role access, calendar synchronization, and service recovery. • The quadrant mix is weighted to Q1 and Q4 because the highest risks are concurrency, authorization, and synchronization. Q3 is smaller at Midterm because execution is out of scope. No equal quota is used. • Context drives choices: four roles drive a role x data decision table; synthetic data only because of data-privacy constraints; Chrome, Edge, and Android only because	Section VIII: Approach; Quadrants Summary Appendix D: Coverage Map Section III: Assumptions and Constraints Section VII: Not To Be Tested

		iOS is unavailable. Each exclusion has a stated reason.	
7	Absence-of-defects fallacy *Required: plan includes business and user validation, not only technical defect detection.*	• The Acceptance Test Plan validates the Patient, Doctor, and Clinic Admin/Staff journeys (UAT-1 to UAT-3) and a privacy check (UAT-4). • Acceptance pass criteria are user-centered, for example the Patient journey is completed without assistance and the recovery prompt is clear to the Doctor. • Q2 acceptance-criteria tests, Q3 exploratory sessions, and a usability walkthrough of the booking and recovery screens add business-facing validation. UAT-1 and UAT-2 must pass before acceptance exit. Full UAT execution is planned but not required at Midterm.	Acceptance Test Plan: Sections IV, VIII, IX Section VIII: Quadrants Summary Appendix D: Q2 and Q3 Section VII: Full UAT execution

VII. Features Not To Be Tested

Feature / Area	Reason for Exclusion
Test execution results as principle evidence	Scope: no dynamic execution at Midterm. No execution evidence, screenshots, or defect counts are claimed (Student Guide Section 12).
Evidence from actual defect clusters (Principle 4)	Context: no defects exist yet. Cluster evidence is by risk score now and is updated after execution.
Technical correctness of each test condition	Scope: covered by the level test plans and Appendix B, not by this review.
Ranking or scoring the principles against each other	Scope: each principle is only judged to pass or fail against its criteria.

VIII. Approach

The approach is a checklist-based static review of the MTP. Static review sits outside the testing quadrant count, as the Student Guide requires. It is done by someone other than the writer of each evidence statement.

1. List the evidence statement for each principle (Section VI).
2. Open each cited section or appendix and confirm it says what the statement claims.
3. Check that the statement is MediLynk-specific, meaning it names real REQ, PR, TI, or ER identifiers.
4. Search the MTP and all four level test plans for “defect-free”, “no defects”, or similar claims.
5. Record pass or fail per item (Section IX). Fix failed items and review them again.
6. Repeat the check after any renumbering of sections or any new MTP version.

Entry criteria: MTP v1.0 text is stable; Appendices A to D are complete; every group member has access to the current DOCX.

Test Tools

Tool	**Purpose**
Microsoft Word (Find, Navigation Pane, Comments)	Locate cited sections, search for prohibited claims, and record review comments
Student Guide Appendix E template and rubric criterion 8	Checklist for required evidence per principle
Jira / Trello	Track review findings and fixes

IX. Item Pass/Fail Criteria

Item	Pass Criteria	Fail Criteria
EP-01	No statement claims testing proves absence of defects. Exit criteria include residual risk and a Test Completion Report.	“Defect-free” or “no defects” appears as a goal or criterion.
EP-02	Scope, exclusions, techniques, and coverage goals are explicit and cited.	Claims to test everything, or exclusions are unexplained.
EP-03	Static review and risk analysis are dated before test design, and findings are marked resolved in v1.0.	Static review is missing or happened after test design.
EP-04	High risks are tested first and deeper, and cluster monitoring is defined.	All areas get the same depth with no risk basis.
EP-05	Regression review and data refresh tasks exist in the plan.	The same tests and data are assumed valid forever.
EP-06	Approach cites Scenario D features, roles, and constraints.	Text is generic and could describe any system.
EP-07	UAT scenarios and Q3 usability work are planned with real user roles.	Only technical tests are planned.

Exit Criteria

- All seven items EP-01 to EP-07 pass.

- Every “Where the Group Shows It” reference points to a section or appendix that exists in the current MTP.
- A reviewer other than the writer has checked every row.
- Any unresolved finding is listed with its reason.

Passing this review shows that the principles are evidenced in the plan. It does not show that the MTP or MediLynk is free of defects.

X. Suspension Criteria and Resumption Requirements

Suspension criteria: the review stops if

- Section numbers or the structure of the MTP change, so cited references become invalid.
- A statement claiming the system is defect-free is found in the MTP.
- The MTP version under review is replaced by a newer version.

Resumption requirements:

- Matrix references are updated to match the current MTP.
- The MTP version under review is confirmed by the Project Manager.
- The Test Manager approves the resumption.

XI. Test Deliverables

- Seven Testing Principles Evidence Matrix (Section VI of this appendix)
- Review checklist with pass or fail result per item EP-01 to EP-07
- List of review findings and their fixes

- Revision History entry in the MTP for any change made because of this review

XII. Testing Task

No.	Testing Task	Owner (role)
1	Draft one evidence statement per principle from MTP v1.0	QA Engineer
2	Verify that each cited section or appendix says what the matrix claims, in the owner's own area	Test Analyst / Technical Test Analyst / Integration Test Engineer / Test Manager
3	Search the MTP and level plans for defect-free claims	QA Engineer
4	Record pass or fail, fix failed rows, and review again	Project Manager
5	Update references after any renumbering and add the revision history entry	Project Manager
6	Final check against Section IX exit criteria and report	Test Manager

XIII. Environmental Needs

Requirement	Specification
Documents	Current MTP v1.0 in DOCX and PDF; IT302 Midterm Project Student Guide

Software	Microsoft Word (or compatible) with Find and Comments; PDF reader
Workstations	Laptop or desktop with Windows 10/11 and shared access to the group's files
Tracking	Jira or Trello board for review findings
Test data	None. This is a document review with no system execution.

XIV. Responsibilities

Role	**Assigned Member**	**Responsibility in This Review**
QA Engineer	Tamayo, Jonie R.	Draft evidence statements; search for prohibited claims; lead the static review
Project Manager	Maranan, Aryana Rizanne N.	Consolidate the matrix; update references and the revision history
Test Analyst	Aduna, Franz Juris U.	Verify claims about requirements, test design, and techniques (Sections VI and VIII; Appendix B)
Technical Test Analyst	Marasigan, Kylie Denise D.G.	Verify claims about quadrants, exclusions, and environment (Sections VII and VIII; Appendix D)

Integration Test Engineer	Estolloso, Cielo Marie S.	Verify claims about test data and regression (Section XII; IR-03)
Test Manager	Salvador, Mary Pauleen C.	Verify claims about risks and exit criteria (Sections V and IX; Appendix A); approve resumption; final report

XV. Staffing and Training Needs

Reviewers need working knowledge of the seven ISTQB principles (CTFL Sec. 1.3) and the ability to read the MTP and trace identifiers.

Training: a short walkthrough of the Student Guide Section 4 table and rubric criterion 8, so every member can explain how each principle influenced the plan (Student Guide Section 12).

XVI. Schedule

Dates follow the MTP schedule in Section XVI. Effort is estimated in person-hours.

Period	Activity	Responsible	Effort (person-hours)
Sep 28	Draft the evidence matrix from MTP v1.0	Tamayo / Maranan	3
Sep 29	Peer static review of the matrix; fix findings; update revision history	All members	3
Oct 1	Final reference check against the restructured MTP	Tamayo / Maranan	1

XVII. Risks and Contingencies

ID	Risk	Mitigation	Contingency
AE-01	Evidence statements are generic and could fit any system	Name MediLynk REQ, PR, TI, and ER identifiers in each statement	Reviewer returns the statement for rewriting before baseline
AE-02	Cited section numbers become wrong after the MTP is edited	Re-check references after every edit and each new version	Update the affected rows and review them again
AE-03	Evidence overstates what the plan does,	Reviewer checks each claim against the cited text;	Reword or remove the claim and log a review finding

	such as implying tests were already executed	no execution results are claimed	

XII. Remaining Test Tasks

After the Master Test Plan has been reviewed and approved, the testing team will complete the remaining activities required to evaluate the system. These activities will be performed systematically to ensure that important functions and integration points are properly verified.

Remaining Test Task	Description	Owner (role)
1. Finalization of Test Cases	Review and finalize test cases based on approved requirements, identified risks, and feedback from the static review.	Test Analyst
2. Test Data Preparation	Prepare sample patient profiles, healthcare personnel accounts, appointment schedules, and reminder records.	Integration Test Engineer
3. Test Environment Configuration	Set up the required hardware, software, database, network connection, testing tools, and system build.	Technical Test Analyst
4. Test Execution	Execute the approved test cases and check whether actual results meet expected results.	All testers
5. Defect Identification and Reporting	Record defects and document severity, priority, reproduction steps, and possible effects.	All testers
6. Defect Retesting	Repeat failed test cases after developers apply corrections.	Test Analyst

Remaining Test Task	Description	Owner (role)
7. Regression Testing	Recheck previously working features after changes to detect new defects.	Test Analyst / Integration Test Engineer
8. Test Case and Data Review	Review test cases and data for outdated, duplicated, incomplete, or ineffective material; refresh tests and regenerate date-sensitive appointment and reminder data after each fix cycle.	Test Manager
9. Testing Progress Monitoring	Monitor execution, defect status, coverage, and completion of assigned responsibilities.	Test Manager
10. Test Completion Reporting	Prepare the completion report with results, unresolved issues, residual risks, limitations, and recommendations.	Test Manager

XIII. Environmental Needs

The testing environment must provide the hardware, software, network access, test accounts, data, and tools needed to evaluate the system. The requirements below are proposed and should be confirmed against the technologies selected by the development team.

A. Hardware, Software, and Tools

Requirement	Specification	Purpose
Workstation	Laptop/desktop, Intel Core i5 or equivalent, 8 GB RAM, 256 GB free storage; Windows 10/11	Run the system, browsers, database, and tools
Web browsers	Google Chrome and Microsoft Edge	Test the React web application
Mobile	Android device or emulator (iOS not available)	Test the Flutter mobile application
Application server	Node.js/Express (MVC) build	System under test
Database	MySQL (InnoDB) with the six data-tier tables; needed for transactions and the unique slot constraint	Verify storage, unique slot constraint, and consistency
Video simulator	Mock video-conferencing simulator	Simulate video sessions, outage, and dropped calls
Notification services	Email/SMS/Push sandbox accounts, or mock notification service	Verify reminder dispatch (TI-06, TI-10)
Test tools	Postman, JMeter, OWASP ZAP, Jira/Trello, VS Code	API, concurrency, security, tracking, editing

B. Network and Integration

Requirement	Specific Need
Internet connection	Stable connection for sandbox providers and hosted services
Local network	If the application is hosted locally, a network linking the app, database, and test devices
API / external services	Reachable REST endpoints (TI-14) and provider sandboxes or mocks (TI-10)

C. Test Accounts and Test Data

Requirement	Description
Patient accounts	Valid and invalid login data; at least two patients to test cross-patient access
Doctor accounts	Doctor/Healthcare Provider accounts with schedules
Clinic Admin/Staff accounts	Separate from System Administrator; needed for REQ-02 and REQ-05 decision tables

Requirement	Description
System Administrator accounts	Platform-level accounts
Appointment data	Available, occupied, cancelled, and expired slots; buffer-boundary cases
Reminder data	Appointments at different dates, times, and reminder statuses
Invalid and boundary data	Missing fields, invalid formats, duplicates, conflicting schedules
Protected test data	Synthetic or anonymized data only

XIV. Staffing and Training Needs

	***PROJECT MANAGER*** **Maranan, Aryana Rizanne N.** The Project Manager is responsible for overseeing and coordinating the overall testing activities of the project. This role includes coordinating the Master Test Plan (MTP), assigning testing tasks to team members, monitoring testing progress and schedules, coordinating communication among the team, and consolidating test results and other testing outputs. The Project Manager also monitors testing risks and issues to help ensure that all testing activities are completed properly and on time. This role requires skills in test planning and management, project coordination, communication, task delegation, monitoring, and documentation.
	***TEST MANAGER*** **Salvador, Mary Pauleen C.** The Test Manager is responsible for managing and monitoring the overall testing process to ensure that testing activities are properly scheduled, executed, and documented. This role includes identifying and analyzing testing risks, managing the testing schedule, monitoring the progress and results of test activities, and preparing testing reports. The Test Manager also evaluates significant testing issues and approves the suspension or resumption of testing when necessary. This role requires skills in risk analysis, test scheduling, monitoring, decision-making, communication, and reporting.

	***TEST ANALYST*** **Aduna, Franz Juris U.** The Test Analyst is responsible for analyzing the system requirements and determining the appropriate testing activities needed to verify that the system functions as expected. This role includes developing test scenarios and test cases, preparing test data, defining expected results, executing tests, documenting actual results, and reporting identified defects. The Test Analyst also supports retesting after defects are fixed and ensures that test cases are properly aligned with the system requirements. This role requires skills in requirements analysis, test design, test case development, test execution, defect reporting, and documentation.
	***TECHNICAL TEST ANALYST*** **Marasigan, Kylie Denise D.G.** The Technical Test Analyst is responsible for evaluating the technical aspects of the system, particularly its security, performance, compatibility, and test environment. This role includes conducting security and authorization testing, checking the system's performance under different workloads and concurrent users, and performing compatibility testing across supported devices, browsers, and environments. The Technical Test Analyst also assists in setting up and configuring the test environment and uses tools such as JMeter for performance testing and OWASP ZAP for security testing. This role requires skills in security and authorization testing, performance and concurrency testing, compatibility testing, test environment setup, and the use of technical testing tools.

	***QUALITY ASSURANCE ENGINEER*** **Tamayo, Jonie R.** The Quality Assurance Engineer is responsible for ensuring that the system and testing activities meet the required quality standards and project requirements. This role includes conducting static reviews of requirements and documentation, checking quality and compliance, and performing usability and exploratory testing to identify issues that may not be found through standard test cases. The Quality Assurance Engineer also reviews testing documents and results to ensure accuracy, completeness, and proper traceability between requirements, test cases, and identified defects. This role requires skills in quality assurance, static review, compliance checking, usability and exploratory testing, documentation, and requirements traceability.
	***INTEGRATION TESTING*** **Estolloso, Cielo Marie S.** The Integration Tester is responsible for verifying that the different components and services of the system communicate and exchange data correctly. This role includes conducting API and integration testing, checking database interactions, preparing test data, and verifying the accuracy and consistency of information transferred between the booking system, database, calendar service, and reminder services. The Integration Tester also identifies integration errors, checks how the system handles failed requests, and documents issues encountered during testing. This role requires skills in API and integration testing, database testing, test data preparation, data flow verification, and defect reporting.

XV. Responsibilities

Role	Assigned Member	Main Responsibilities
Project Manager	Maranan, Aryana Rizanne N.	Coordinate MTP, assign tasks, monitor progress, consolidate outputs.
Test Manager	Salvador, Mary Pauleen C.	Manage testing activities, priorities, schedule, monitoring.
Test Analyst	Aduna, Franz Juris U.	Analyze requirements, identify test conditions, design functional tests.
Technical Test Analyst	Marasigan, Kylie Denise D.G.	Handle technical testing concerns (security, performance, compatibility).
Quality Assurance Engineer	Tamayo, Jonie R.	Conduct quality reviews, static reviews, compliance checks.
Integration Test Engineer/Tester	Estolloso, Cielo Marie S.	Focus on integration points, interfaces, data flow.

Activity	Owner
Planning	Maranan
Risk analysis	Salvador
Static review	Tamayo
Test design	Aduna
Test data preparation (TR-03)	Estolloso
Environment preparation	Marasigan
Documentation and review	Maranan / Tamayo
Monitoring and reporting	Salvador

XVI. Schedule

Day	Date	Main Activity	Responsible	Effort (person-hours)	Expected Output
1	Sep 15	Analyze Scenario D	Aduna	4	REQ-01 to REQ-10
2	Sep 16	Early/static review	Tamayo	4	Appendix C findings; MTP v0.1 issues
3	Sep 17	Product and project/test risks	Salvador	4	Appendix A
4	Sep 18-19	Objectives, scope, items, features	Aduna / Maranan	8	Sections I-VII
5	Sep 21	Priorities and quadrants map	Marasigan	4	Appendix D
6	Sep 22	Test approach / strategy	Marasigan	4	Section VIII
7	Sep 23	Pass/fail, entry/exit, suspension	Marasigan	4	Sections IX-X
8	Sep 24-25	Deliverables, tasks, environment, roles, contingencies	Estolloso / Maranan	8	Sections XI-XVII

Day	Date	Main Activity	Responsible	Effort (person-hours)	Expected Output
9	Sep 28	Approvals, glossary, principles evidence; RTM	Tamayo / Maranan	8	Sections XVIII-XIX; Appendices B, E
10	Sep 29	Peer static review, revise, baseline, submit	All	12	MTP Version 1.0

XVII. Planning Risks and Contingencies

ID	Risk	L	I	Score	Mitigation	Contingency
TR-01	Requirements for schedule overlap and cancellation rules stay unclear	3	3	9	Clarify during static review (ER-06)	Document as an open assumption and flag for instructor review

ID	Risk	L	I	Score	Mitigation	Contingency
TR-02	Email/SMS/Push sandbox unavailable or its free-tier quota is exhausted	4	3	12	Use the mock notification service; confirm sandbox access before execution	Rely on the mock. Real third-party delivery (TI-10) stays unverified and is recorded as residual risk
TR-03	Test data for schedules and patient records is missing or insufficient	2	2	4	Prepare sample data sets in advance; assign one member to maintain test data	Generate minimal data manually before execution
TR-04	Limited meeting time delays planning tasks	2	3	6	Set internal deadlines ahead of the official schedule; track progress each meeting	Reassign to a member with lighter workload

XVIII. Approvals

Role	Name	Signature	Date	Version	Status
Prepared by	Maranan, Aryana Rizanne N. (Project Manager)		October 4, 2026	1.0	Prepared
Reviewed by	Tamayo, Jonie R. (Quality Assurance Engineer)		October 4, 2026	1.0	Reviewed
Approved by	Engr. Evangeline F. Gonzales			1.0	Pending approval

BULACAN STATE UNIVERSITY

College of Information and Communication Technology

Pimentel Hall, City of Malolos

Dear Ms. Maranan,

We are pleased to confirm your appointment as the **Coordinator** for our **Clinic Teleconsultation Booking and Reminder System** project in **Systems Integration and Architecture**.

Our group's objective is to successfully design and develop a system that enables users to book clinic teleconsultations and receive automated appointment reminders. Each member plays a vital role in ensuring the quality, organization, and successful completion of the project.

As the **Coordinator**, you are expected to fulfill the following responsibilities:

You are responsible for keeping the group on task and managing time.

Each member is expected to actively participate, communicate effectively, complete assigned tasks on time, and contribute to achieving the group's objective.

If you fail to perform your assigned responsibilities **without any valid reason**, the following sanctions will apply:

- **First Offense** – Counseling and discussion with the group.
- **Second Offense** – Formal warning and notice of possible removal from the group.
- **Third Offense** – Subject to removal from the group.

By signing this appointment, you acknowledge that you fully understand and accept the responsibilities, expectations, and conditions assigned to your role.

This appointment becomes effective immediately and shall remain valid until the successful completion of this semester's Systems Integration and Architecture project.

Prepared by:

Jonie R. Tamayo

Witnessed by:

Kylie Denise D.G. Marasigan

Noted by:

Engr. Evangeline F. Gonzales

Conformed by:

Aryana Rizanne N. Maranan

BULACAN STATE UNIVERSITY

College of Information and Communication Technology

Pimentel Hall, City of Malolos

Dear Ms. Salvador,

We are pleased to confirm your appointment as the **System Analyst** for our **Clinic Teleconsultation Booking and Reminder System** project in **Systems Integration and Architecture**.

Our group's objective is to successfully design and develop a system that enables users to book clinic teleconsultations and receive automated appointment reminders. Each member plays a vital role in ensuring the quality, organization, and successful completion of the project.

As the **System Analyst**, you are expected to fulfill the following responsibilities:

You are responsible for identifying the system's users, features, and components. You will ensure that all functional and non-functional requirements are accurately defined, consistent, and well-documented for the development phase.

Each member is expected to actively participate, communicate effectively, complete assigned tasks on time, and contribute to achieving the group's objective.

If you fail to perform your assigned responsibilities **without any valid reason**, the following sanctions will apply:

- **First Offense** – Counseling and discussion with the group.
- **Second Offense** – Formal warning and notice of possible removal from the group.
- **Third Offense** – Subject to removal from the group.

By signing this appointment, you acknowledge that you fully understand and accept the responsibilities, expectations, and conditions assigned to your role.

This appointment becomes effective immediately and shall remain valid until the successful completion of this semester's Systems Integration and Architecture project.

Prepared by:

Jonie R. Tamayo

Witnessed by:

Cielo Marie S. Estolloso

Noted by:

Eng. Evangeline F Gonzales

Conformed by:

Mary Pauleen C. Salvador

BULACAN STATE UNIVERSITY

College of Information and Communication Technology

Pimentel Hall, City of Malolos

Dear Ms. Marasigan,

We are pleased to confirm your appointment as the **Architecture Mapper** for our **Clinic Teleconsultation Booking and Reminder System** project in **Systems Integration and Architecture**.

Our group's objective is to successfully design and develop a system that enables users to book clinic teleconsultations and receive automated appointment reminders. Each member plays a vital role in ensuring the quality, organization, and successful completion of the project.

As the **Architecture Mapper**, you are expected to fulfill the following responsibilities:

You are responsible for connecting components, architecture, and integration choices. You will ensure that the system's structural design is cohesive, logical, and that all technical integrations are accurately mapped out and documented before submission.

Each member is expected to actively participate, communicate effectively, complete assigned tasks on time, and contribute to achieving the group's objective.

If you fail to perform your assigned responsibilities **without any valid reason**, the following sanctions will apply:

- **First Offense** – Counseling and discussion with the group.
- **Second Offense** – Formal warning and notice of possible removal from the group.
- **Third Offense** – Subject to removal from the group.

By signing this appointment, you acknowledge that you fully understand and accept the responsibilities, expectations, and conditions assigned to your role.

This appointment becomes effective immediately and shall remain valid until the successful completion of this semester's Systems Integration and Architecture project.

Prepared by:

Jonie R. Tamayo

Witnessed by:

Jonie R. Tamayo

Noted by:

Eng. Evangeline F Gonzales

Conformed by:

Kylie Denise D.G. Marasigan

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Dear Mr. Aduna,

We are pleased to confirm your appointment as the **Testing Analyst** for our **Clinic Teleconsultation Booking and Reminder System** project in **Systems Integration and Architecture**.

Our group's objective is to successfully design and develop a system that enables users to book clinic teleconsultations and receive automated appointment reminders. Each member plays a vital role in ensuring the quality, organization, and successful completion of the project.

As the **Testing Analyst**, you are expected to fulfill the following responsibilities:

You are responsible for tracing defects, risks, and test ideas. You will ensure that the system is thoroughly evaluated for potential issues, and that all testing procedures are properly executed and documented to maintain system quality before submission.

Each member is expected to actively participate, communicate effectively, complete assigned tasks on time, and contribute to achieving the group's objective.

If you fail to perform your assigned responsibilities **without any valid reason**, the following sanctions will apply:

- **First Offense** – Counseling and discussion with the group.
- **Second Offense** – Formal warning and notice of possible removal from the group.
- **Third Offense** – Subject to removal from the group.

By signing this appointment, you acknowledge that you fully understand and accept the responsibilities, expectations, and conditions assigned to your role.

This appointment becomes effective immediately and shall remain valid until the successful completion of this semester's Systems Integration and Architecture project.

Prepared by:

Jonie R. Tamayo

Witnessed by:

Aryana Rizanne N. Maranan

Noted by:

Eng. Evangeline F Gonzales

Conformed by:

Franz Juris U. Aduna

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Dear Mr. Tamayo,

We are pleased to confirm your appointment as the **Documentation Lead** for our **Clinic Teleconsultation Booking and Reminder System** project in **Systems Integration and Architecture**.

Our group's objective is to successfully design and develop a system that enables users to book clinic teleconsultations and receive automated appointment reminders. Each member plays a vital role in ensuring the quality, organization, and successful completion of the project.

As the **Documentation Lead**, you are expected to fulfill the following responsibilities:

You are responsible for checking the clarity and completeness of individual outputs. You will ensure that every member's work is well-organized, consistent, and properly compiled into the final document before submission.

Each member is expected to actively participate, communicate effectively, complete assigned tasks on time, and contribute to achieving the group's objective.

If you fail to perform your assigned responsibilities **without any valid reason**, the following sanctions will apply:

- **First Offense** – Counseling and discussion with the group.
- **Second Offense** – Formal warning and notice of possible removal from the group.
- **Third Offense** – Subject to removal from the group.

By signing this appointment, you acknowledge that you fully understand and accept the responsibilities, expectations, and conditions assigned to your role.

This appointment becomes effective immediately and shall remain valid until the successful completion of this semester's Systems Integration and Architecture project.

Prepared by:

Aryana Rizanne N. Maranan

Witnessed by:

Franz Juris U. Aduna

Noted by:

Eng. Evangeline F Gonzales

Conformed by:

Jonie R. Tamayo

BULACAN STATE UNIVERSITY
College of Information and Communication Technology
Pimentel Hall, City of Malolos

Dear Ms. Estolloso,

We are pleased to confirm your appointment as the **Optional Presenter/Reviewer** for our **Clinic Teleconsultation Booking and Reminder System** project in **Systems Integration and Architecture.**

Our group's objective is to successfully design and develop a system that enables users to book clinic teleconsultations and receive automated appointment reminders. Each member plays a vital role in ensuring the quality, organization, and successful completion of the project.

As the **Optional Presenter/Reviewer**, you are expected to fulfill the following responsibilities:

You are responsible for presenting the group's output and reviewing its accuracy, clarity, and completeness before submission. You will ensure that you are familiar with the project, ready to explain it when required, and provide objective, constructive feedback in coordination with the group.

Each member is expected to actively participate, communicate effectively, complete assigned tasks on time, and contribute to achieving the group's objective.

If you fail to perform your assigned responsibilities **without any valid reason,** the following sanctions will apply:

- **First Offense** – Counseling and discussion with the group.
- **Second Offense** – Formal warning and notice of possible removal from the group.
- **Third Offense** – Subject to removal from the group.

By signing this appointment, you acknowledge that you fully understand and accept the responsibilities, expectations, and conditions assigned to your role.

This appointment becomes effective immediately and shall remain valid until the successful completion of this semester's Systems Integration and Architecture project.

Prepared by:

Jonie R. Tamayo

Witnessed by:

Mary Pauleen C. Salvador

Noted by:

Eng. Evangeline F Gonzales

Conformed by:

Cielo Marie S. Estolloso

XIX. Glossary

API – Application Programming Interface: the set of endpoints through which the clients, internal modules, and third-party providers exchange data (TI-14, TI-10).

Buffer Time – The required gap of 10 minutes between consecutive appointments of the same doctor (ER-06). A booking inside the buffer is rejected.

BVA – Boundary Value Analysis: testing values at the edges of input ranges, such as the buffer edge, past and boundary dates, and the 24 h / 1 h reminder lead times.

Checklist-Based Testing – A technique that verifies a fixed list of points, used here for video-session start and quality (REQ-08) and smoke checks.

Concurrency Testing – Testing what happens when several users act at the same time, such as simultaneous booking requests for one doctor and time slot (PR-01).

Data Masking – Hiding sensitive health fields by default from roles without clearance (REQ-05, PR-06).

Decision Table Testing – A technique that tests combinations of conditions and their outcomes, used for role × data and role × field access rules (PR-02, PR-06).

Entry / Exit Criteria – Entry criteria are the conditions that must be true before dynamic testing starts. Exit criteria are the conditions that show the planned testing is sufficient. Neither requires the system to be defect-free (Sections VIII and IX).

EP – Equivalence Partitioning: dividing inputs into valid and invalid classes and testing one value per class, used for registration fields (PR-04).

ER – Early Review finding ID: an issue found during static review and recorded in Appendix C (ER-01 to ER-08).

Error Guessing – A technique that deliberately tries conflicting or failing actions, such as simultaneous bookings or a simulated outage (PR-01, PR-09).

Exploratory Testing – Unscripted, experience-based testing where test design and execution happen together, used for timing and video sessions (PR-03, PR-05, REQ-08).

ISTQB – International Software Testing Qualifications Board: publisher of the syllabi, testing principles, and testing quadrants used as the basis of this plan.

JMeter – Apache JMeter: a tool for small-scale concurrency and timing tests.

Mock / Sandbox Service – A simulated or test-mode version of a real service, such as the Email, SMS, and Push providers or the video platform, used so tests do not need the real service (TR-02).

MTP – Master Test Plan: this document.

Observer Pattern – A design pattern in which a Notification Manager reacts to appointment events and sends reminders (TI-06).

OWASP ZAP – A security scanning tool used for authorization and data-exposure checks (PR-02, PR-06).

P1 / P2 / P3 – Test priorities derived from risk level. P1 is High risk and tested first; P2 is Medium risk; P3 is Low risk and tested last.

Postman – A tool for sending and checking API requests, used for booking, calendar, and notification endpoints.

PR – Product Risk ID, such as PR-01, scored in Appendix A (PR-01 to PR-09).

Quadrant (Q1-Q4) – ISTQB Testing Quadrants: tests classified by business-facing or technology-facing and by supporting the team or critiquing the product (Appendix D).

RBAC – Role-Based Access Control: access rights based on a user's role (Patient, Doctor, Clinic Admin/Staff, System Administrator).

Regression Testing – Re-running tests after changes to confirm that working features still work.

REQ – Requirement ID, such as REQ-01, for a requirement or test condition in scope (REQ-01 to REQ-10).

Residual Risk – A risk that remains after testing, such as unconfirmed thresholds (ER-06 to ER-08) or unverified third-party delivery (TR-02).

Risk Score – Likelihood × Impact, each rated 1 to 5. High is 16-25, Medium is 8-15, and Low is 1-7.

RTM – Requirements Traceability Matrix: maps requirements to risks, test conditions, levels, techniques, priorities, and quadrants (Appendix B).

Severity – How serious a defect is: Critical (data exposure, an accepted double booking, or a total blocker), High, Medium, or Low (Section IX).

Smoke Testing – Quick checks that confirm the build is stable enough for further testing.

State Transition Testing – A technique that tests valid and invalid moves between states, used for appointment status and calendar sync status (PR-08, PR-05).

Static Review – Examining documents without executing code. It is an early testing activity and is not counted in the quadrants.

Suspension Criteria – The conditions that stop testing. Resumption requirements state what must be true before testing restarts (Section X).

SUT – System Under Test: the MediLynk Teleconsultation System.

Synthetic Data – Artificial test data that contains no real patient information.

TCND – Test Condition ID used in Appendix B (for example, TCND-01).

Teleconsultation – A medical consultation held remotely between a Patient and a Doctor through a video session.

Test Basis – The documents (scenario, blueprint, Appendix C clarifications) from which test conditions are derived.

Test Levels – Component (individual units), Integration (modules and services together), System (the whole system end to end), and Acceptance (validation by intended users).

Testing Principles – The seven ISTQB principles, shown with MediLynk-specific evidence in Appendix E.

TI – Test Item ID, such as TI-02, for a module, service, or component under test (TI-01 to TI-16).

TR – Planning risk ID in Section XVII (TR-01 to TR-04). The level test plans use CR, IR, SR, and AR for their own risks.

UAT – User Acceptance Testing: validation by intended users that the system meets business needs (UAT-1 to UAT-4).

Unique Constraint – A database rule that prevents two records with the same doctor and time slot (InnoDB), protecting against double booking (PR-01).